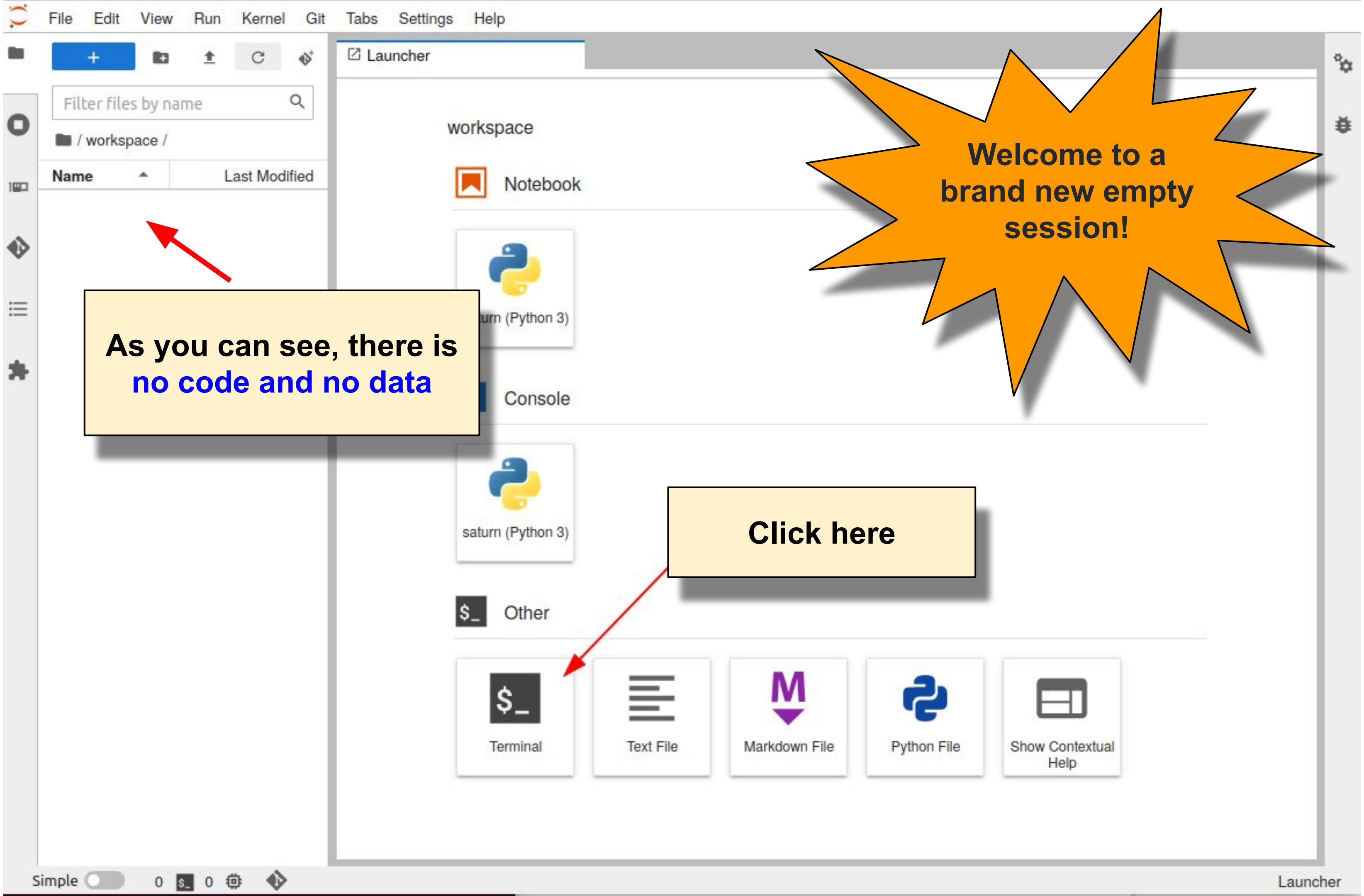
Deep learning for medical imaging school

Hands-on session

Autoencoders

By

Pierre-Marc Jodoin, Thierry Judge

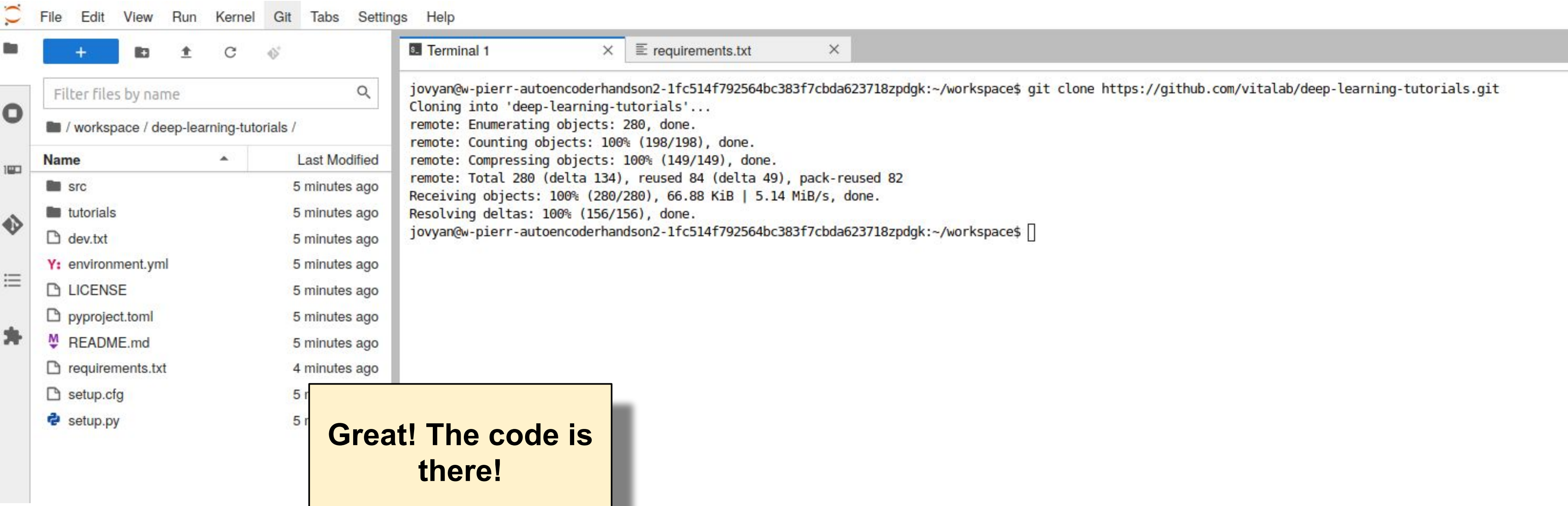

File
Edit
View
Run
Kernel
Git
Tabs
Settings
Help
Launcher
Filter files by name
/ workspace /
Name
Last Modified
workspace
Notebook
urn (Python 3)
Console
saturn (Python 3)
Other
Terminal
Text File
Markdown File
Python File
Show Contextual Help
Simple
Launcher
Welcome to a brand new empty session!
As you can see, there is no code and no data
Click here

In the terminal, type

```
git clone https://github.com/vitalab/deep-learning-tutorials.git
```

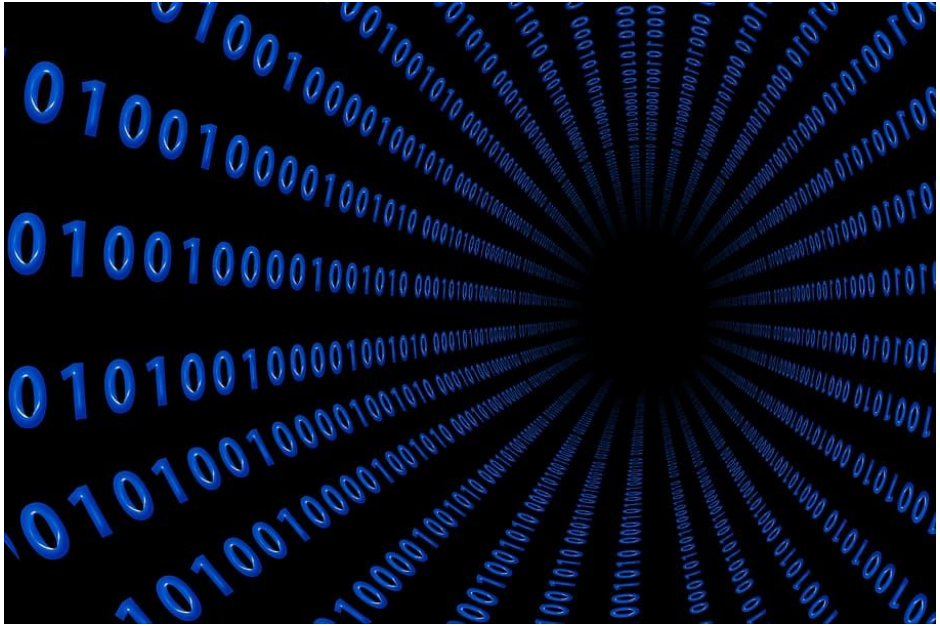
This command allows to **download the code** from a public **gitHub repository**

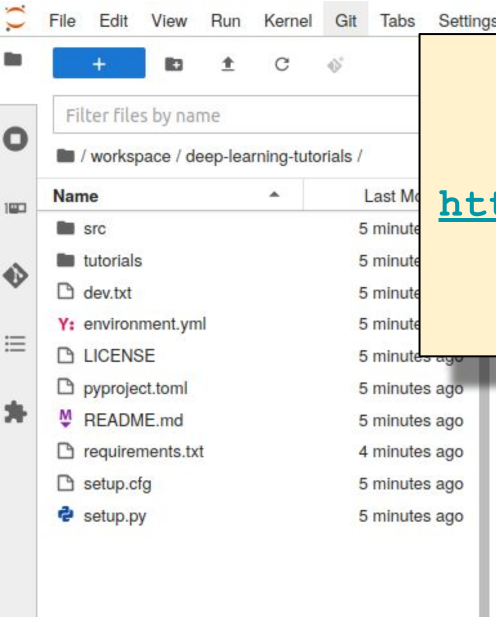
Click here



Great! The code is there!

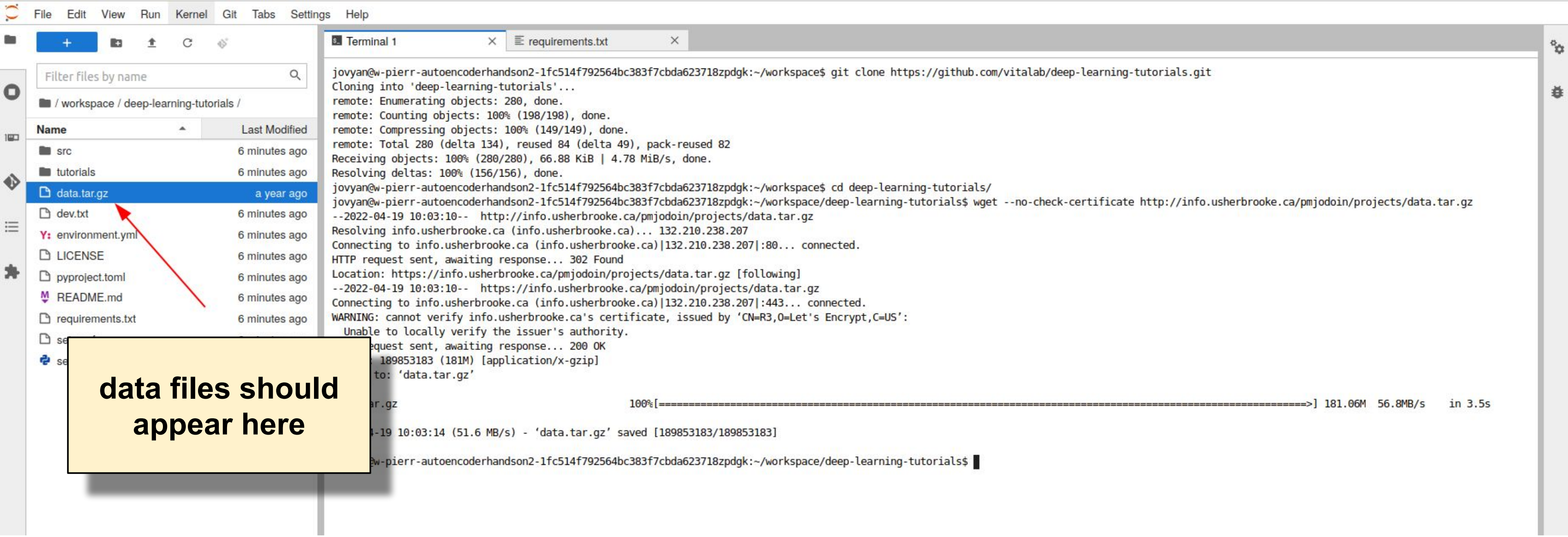
Lets download the data

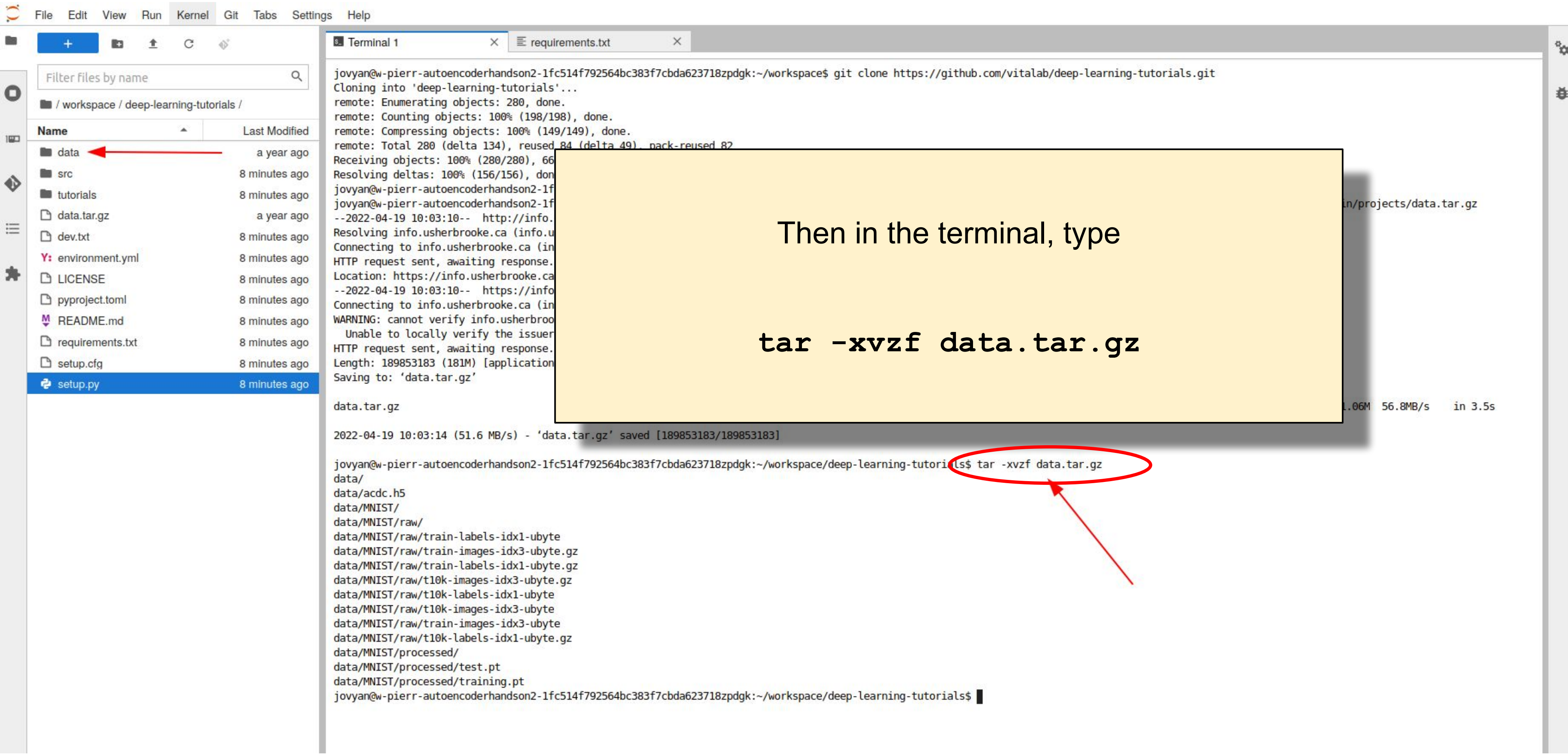


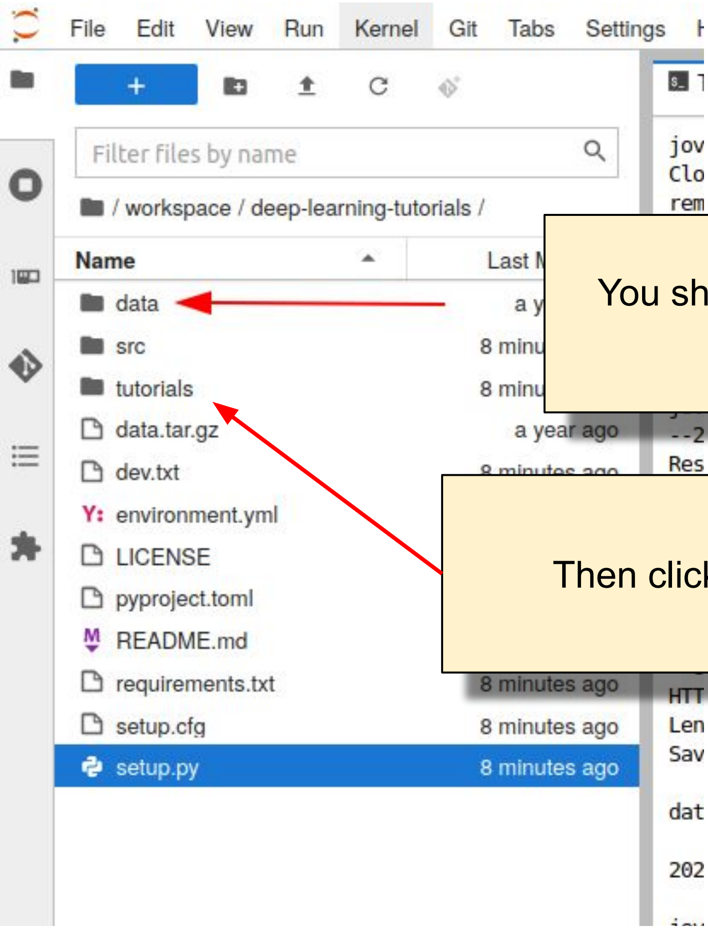


Download the following file on your computer

https://drive.google.com/file/d/1H5pTOYjcSFR6B5GhA0sEPW0wgPVfBq8S/view?usp=sharing

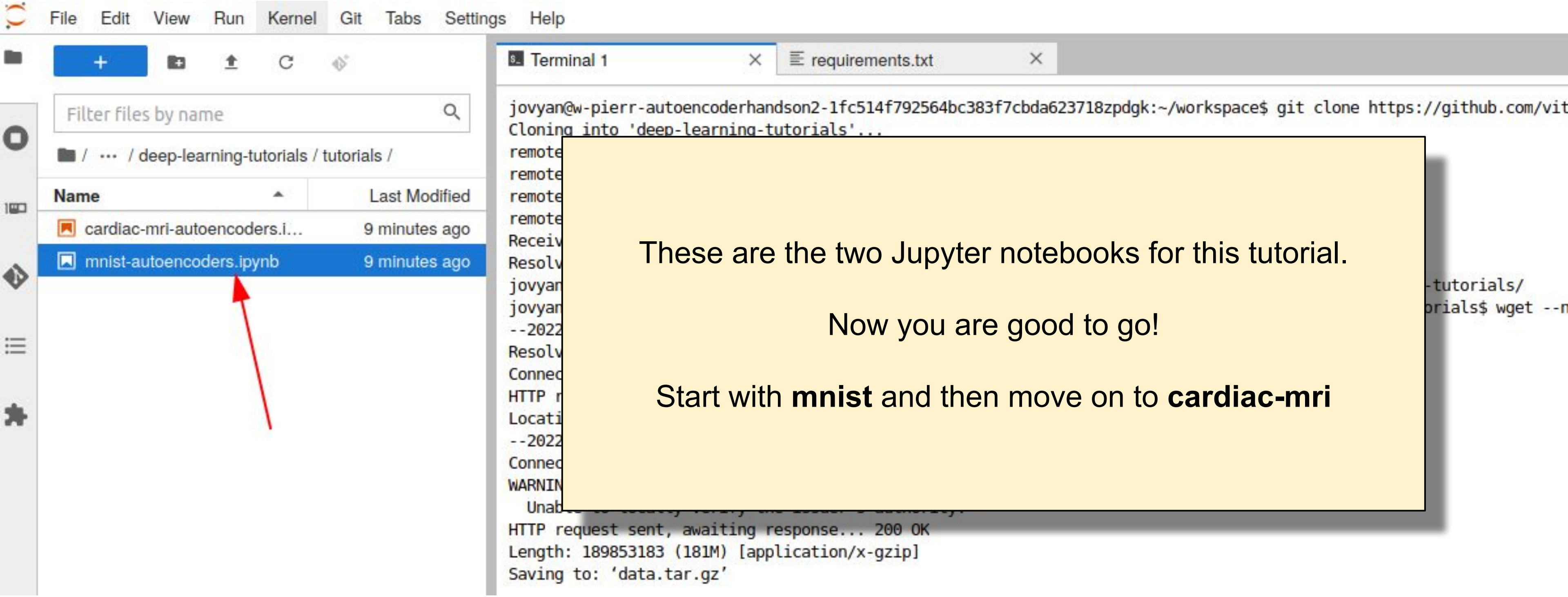

File Edit View Run Kernel Git Tabs Settings Help
Filter files by name
/ workspace / deep-learning-tutorials /
Name
Last Modified
src 6 minutes ago
tutorials 6 minutes ago
data.tar.gz a year ago
dev.txt 6 minutes ago
environment.yml 6 minutes ago
LICENSE 6 minutes ago
pyproject.toml 6 minutes ago
README.md 6 minutes ago
requirements.txt 6 minutes ago
Terminal 1
requirements.txt
jovyan@w-pierr-autoencoderhandson2-1fc514f792564bc383f7cbda623718zpdgk:~/workspace$ git clone https://github.com/vitalab/deep-learning-tutorials.git
Cloning into 'deep-learning-tutorials'...
remote: Enumerating objects: 280, done.
remote: Counting objects: 100% (198/198), done.
remote: Compressing objects: 100% (149/149), done.
remote: Total 280 (delta 134), reused 84 (delta 49), pack-reused 82
Receiving objects: 100% (280/280), 66.88 KiB | 4.78 MiB/s, done.
Resolving deltas: 100% (156/156), done.
jovyan@w-pierr-autoencoderhandson2-1fc514f792564bc383f7cbda623718zpdgk:~/workspace$ cd deep-learning-tutorials/
jovyan@w-pierr-autoencoderhandson2-1fc514f792564bc383f7cbda623718zpdgk:~/workspace/deep-learning-tutorials$ wget --no-check-certificate http://info.usherbrooke.ca/pmjodoin/projects/data.tar.gz
--2022-04-19 10:03:10-- http://info.usherbrooke.ca/pmjodoin/projects/data.tar.gz
Resolving info.usherbrooke.ca (info.usherbrooke.ca)... 132.210.238.207
Connecting to info.usherbrooke.ca (info.usherbrooke.ca)|132.210.238.207|:80... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://info.usherbrooke.ca/pmjodoin/projects/data.tar.gz [following]
--2022-04-19 10:03:10-- https://info.usherbrooke.ca/pmjodoin/projects/data.tar.gz
Connecting to info.usherbrooke.ca (info.usherbrooke.ca)|132.210.238.207|:443... connected.
WARNING: cannot verify info.usherbrooke.ca's certificate, issued by 'CN=R3,O=Let's Encrypt,C=US':
Unable to locally verify the issuer's authority.
equest sent, awaiting response... 200 OK
189853183 (181M) [application/x-gzip]
to: 'data.tar.gz'
r.gz 100%[=====>] 181.06M 56.8MB/s in 3.5s
-19 10:03:14 (51.6 MB/s) - 'data.tar.gz' saved [189853183/189853183]
w-pierr-autoencoderhandson2-1fc514f792564bc383f7cbda623718zpdgk:~/workspace/deep-learning-tutorials$
data files should appear here


File Edit View Run Kernel Git Tabs Settings Help
Terminal 1
requirements.txt
Filter files by name
/ workspace / deep-learning-tutorials /
Name
Last Modified
data a year ago
src 8 minutes ago
tutorials 8 minutes ago
data.tar.gz a year ago
dev.txt 8 minutes ago
environment.yml 8 minutes ago
LICENSE 8 minutes ago
pyproject.toml 8 minutes ago
README.md 8 minutes ago
requirements.txt 8 minutes ago
setup.cfg 8 minutes ago
setup.py 8 minutes ago
jovyan@w-pierr-autoencoderhandson2-1fc514f792564bc383f7cbda623718zpdgk:~/workspace$ git clone https://github.com/vitalab/deep-learning-tutorials.git
Cloning into 'deep-learning-tutorials'...
remote: Enumerating objects: 280, done.
remote: Counting objects: 100% (198/198), done.
remote: Compressing objects: 100% (149/149), done.
Saving to: 'data.tar.gz'
data.tar.gz
2022-04-19 10:03:14 (51.6 MB/s) - 'data.tar.gz' saved [189853183/189853183]
jovyan@w-pierr-autoencoderhandson2-1fc514f792564bc383f7cbda623718zpdgk:~/workspace/deep-learning-tutorials$ tar -xvzf data.tar.gz
data/
data/acdc.h5
data/MNIST/
data/MNIST/raw/
data/MNIST/raw/train-labels-idx1-ubyte
data/MNIST/raw/train-images-idx3-ubyte.gz
data/MNIST/raw/train-labels-idx1-ubyte.gz
data/MNIST/raw/t10k-images-idx3-ubyte.gz
data/MNIST/raw/t10k-labels-idx1-ubyte
data/MNIST/raw/t10k-images-idx3-ubyte
data/MNIST/raw/train-images-idx3-ubyte
data/MNIST/raw/t10k-labels-idx1-ubyte.gz
data/MNIST/processed/
data/MNIST/processed/test.pt
data/MNIST/processed/training.pt
jovyan@w-pierr-autoencoderhandson2-1fc514f792564bc383f7cbda623718zpdgk:~/workspace/deep-learning-tutorials$
Then in the terminal, type
tar -xvzf data.tar.gz



You should have the following **data folder**

Then click on **tutorials**

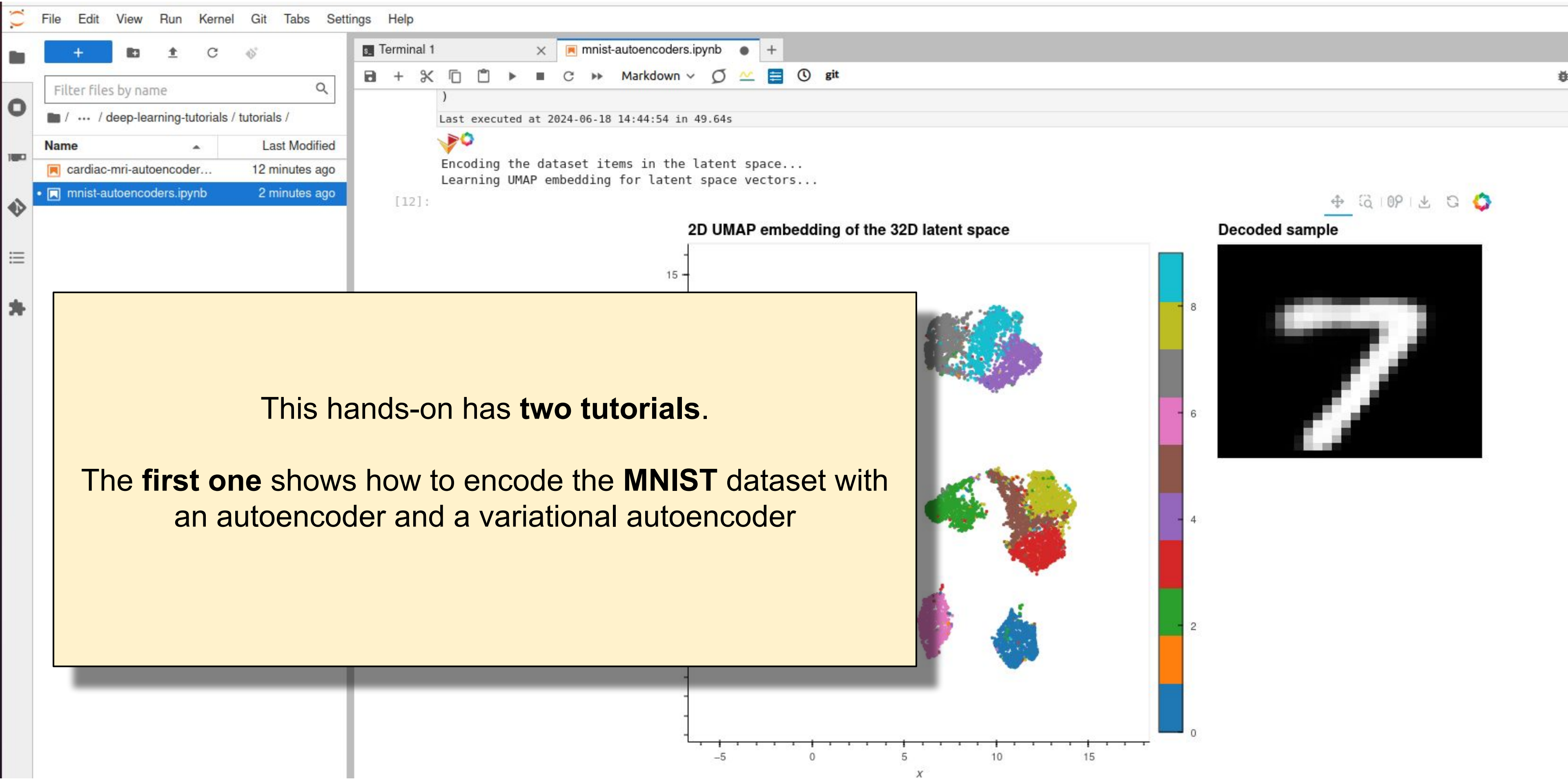


These are the two Jupyter notebooks for this tutorial.

Now you are good to go!

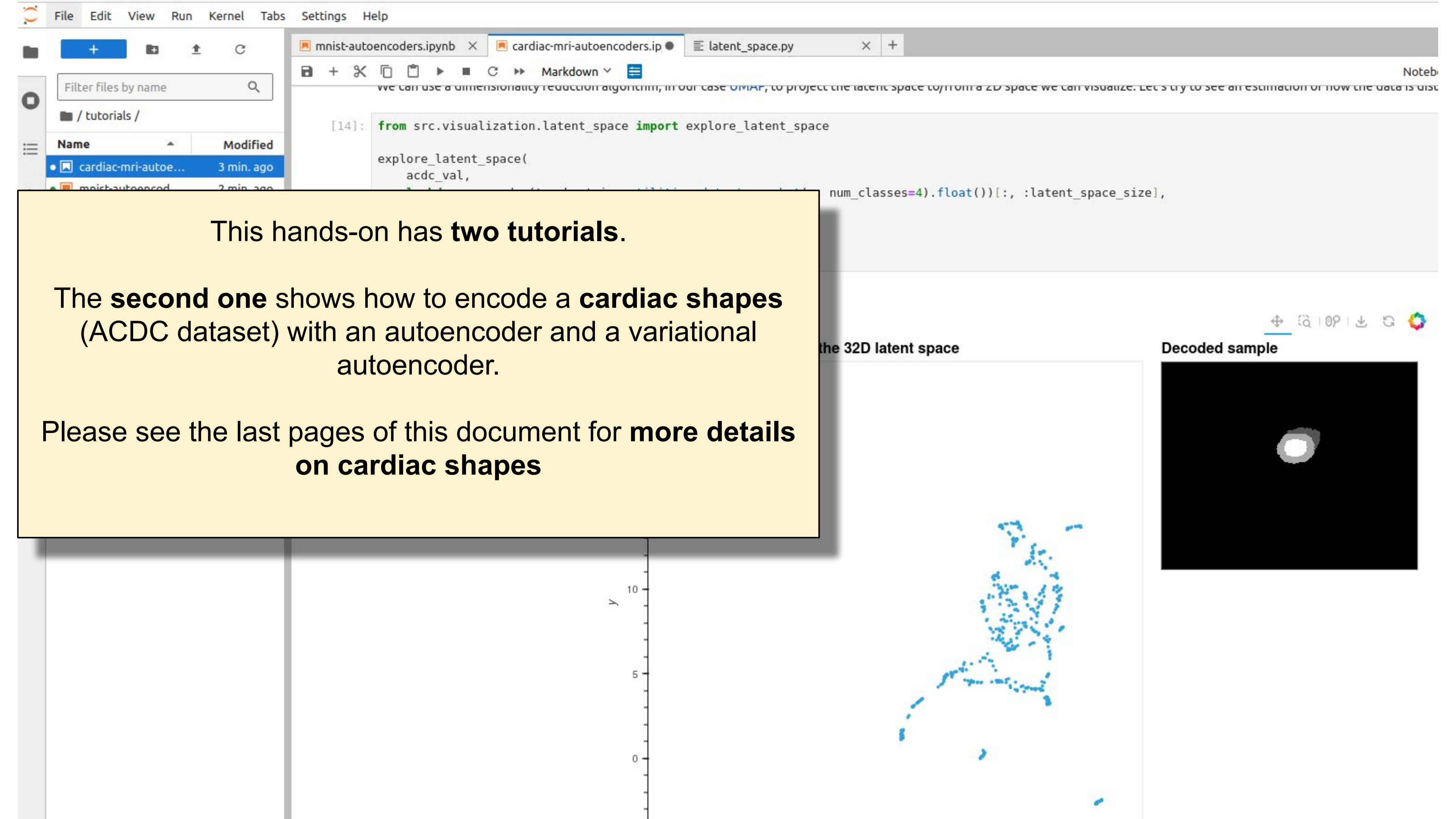
Start with **mnist** and then move on to **cardiac-mri**

Autoencoders'
Hands on



This hands-on has **two tutorials**.

The **first one** shows how to encode the **MNIST** dataset with an autoencoder and a variational autoencoder

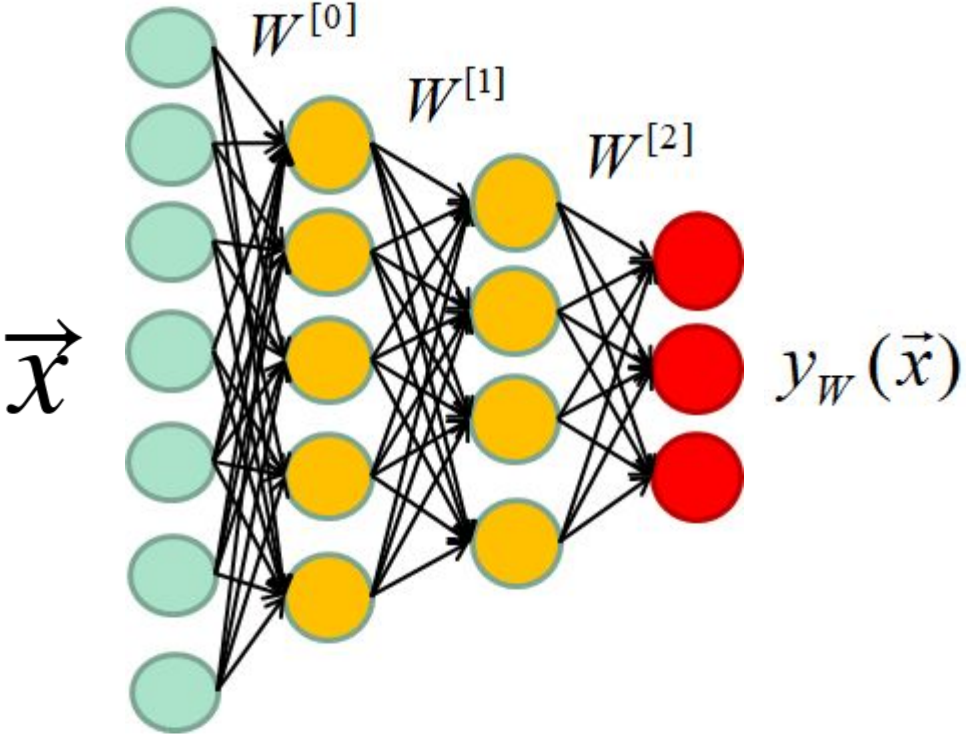


This hands-on has **two tutorials**.

The **second one** shows how to encode a **cardiac shapes** (ACDC dataset) with an autoencoder and a variational autoencoder.

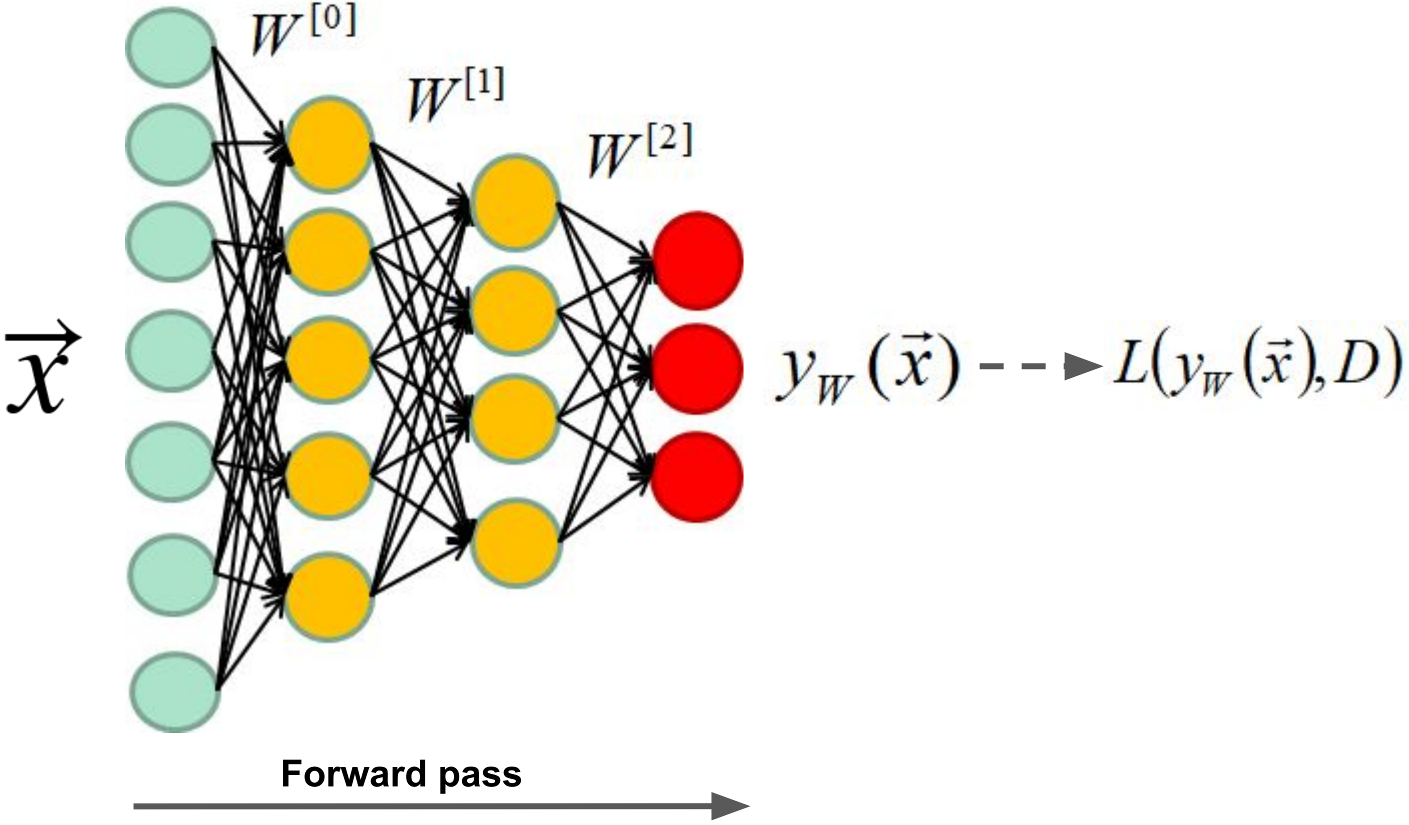
Please see the last pages of this document for **more details on cardiac shapes**

Autoencoders (recap)

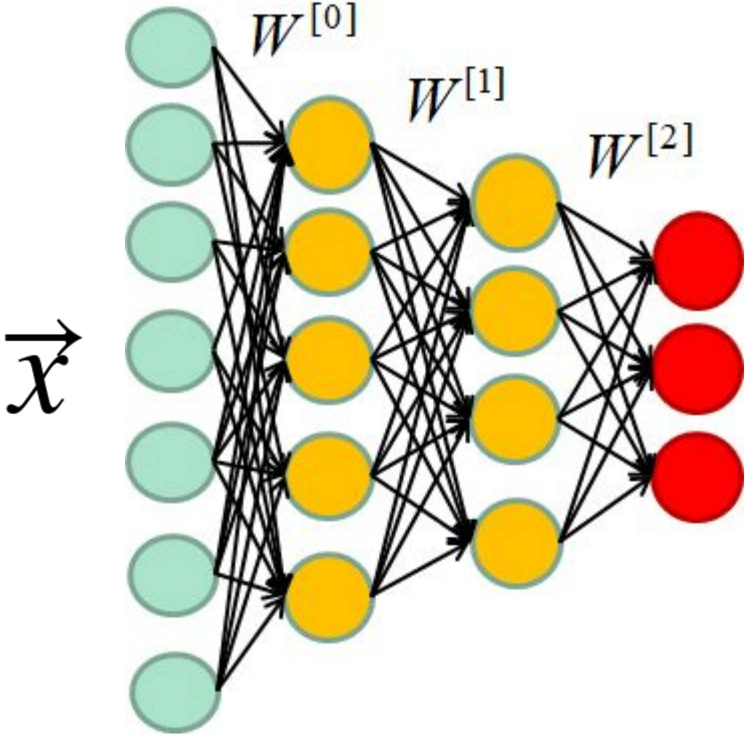


$\vec{x}$
$W^{[0]}$
$W^{[1]}$
$W^{[2]}$
$y_W(\vec{x})$

Annotated dataset: $D = \{(\vec{x}_1, t_1), (\vec{x}_2, t_2), \ldots, (\vec{x}_N, t_N)\}$



Annotated dataset: $D = \{(\vec{x}_1, t_1), (\vec{x}_2, t_2), \ldots, (\vec{x}_N, t_N)\}$

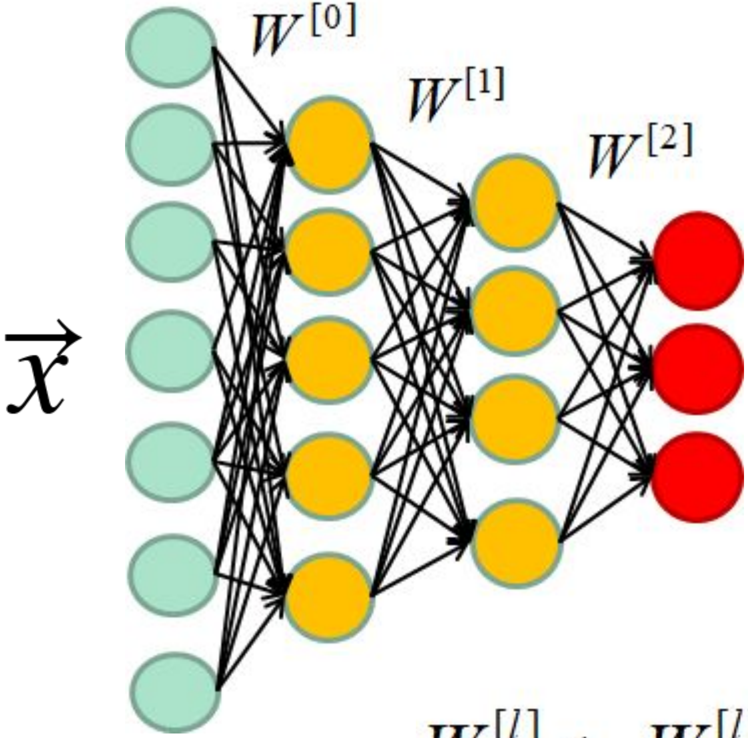


$y_W(\vec{x}) \dashrightarrow L(y_W(\vec{x}), D)$

$$\frac{\partial\left(L\left(y_W(\vec{x})\right)\right)}{\partial W_j^{[l]}}$$

Backward pass

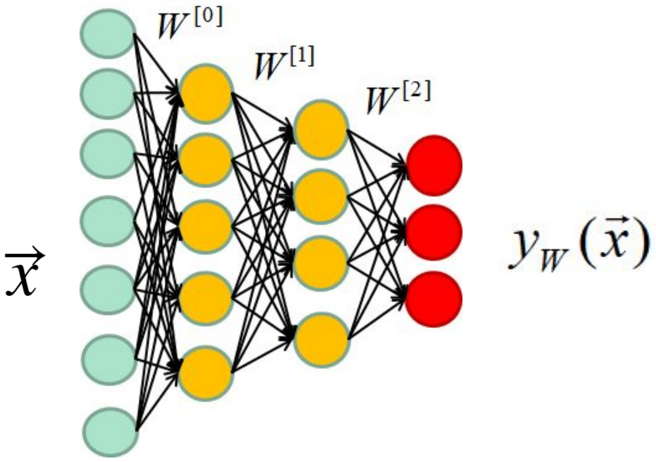
Annotated dataset: $D=\{(\vec{x}_1,t_1),(\vec{x}_2,t_2),\ldots,(\vec{x}_N,t_N)\}$

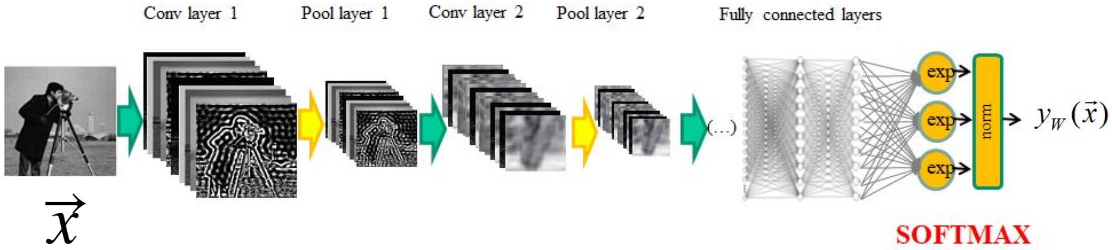


$$y_W(\vec{x}) \dashrightarrow L(y_W(\vec{x}),D)$$

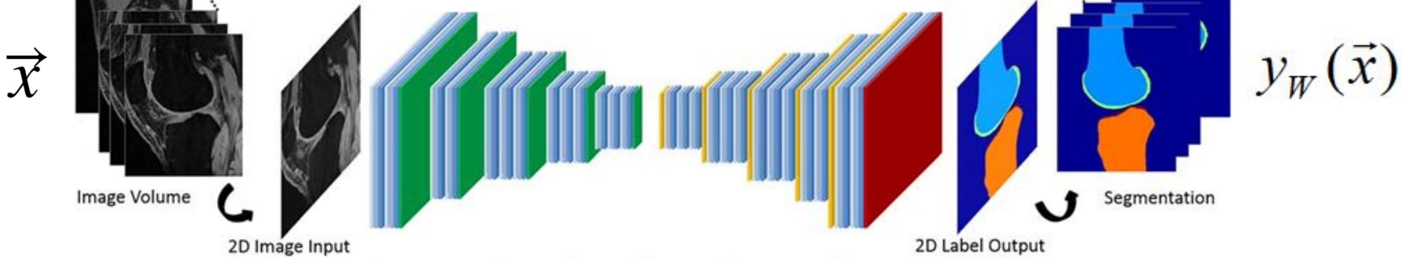
Gradient descent

$$W_j^{[l]} \leftarrow W_j^{[l]} - \eta \frac{\partial(L(y_W(\vec{x})))}{\partial W_j^{[l]}}$$





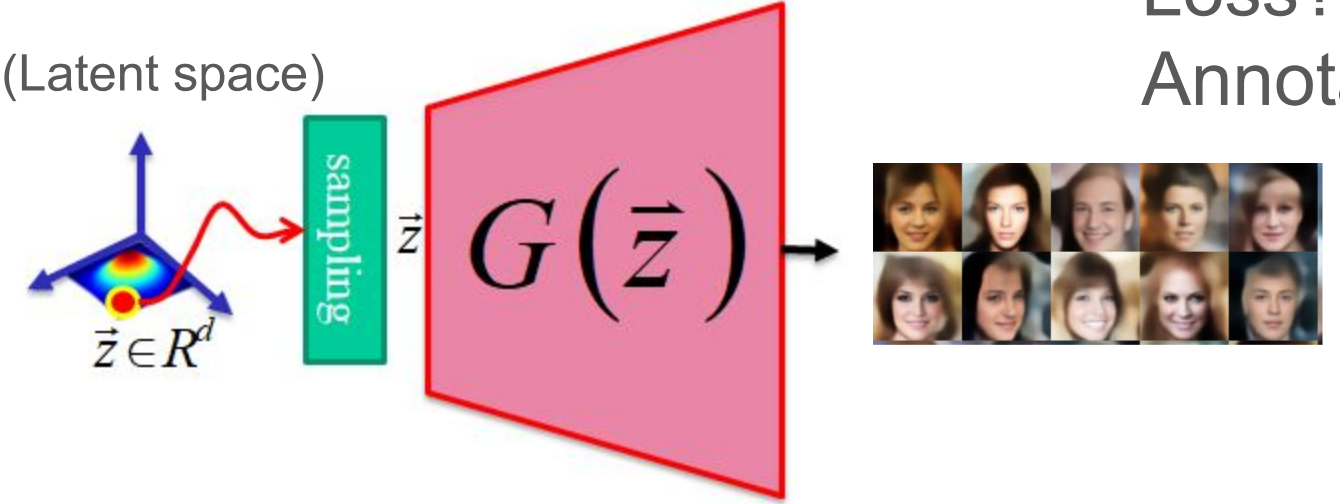
Segmentation CNN



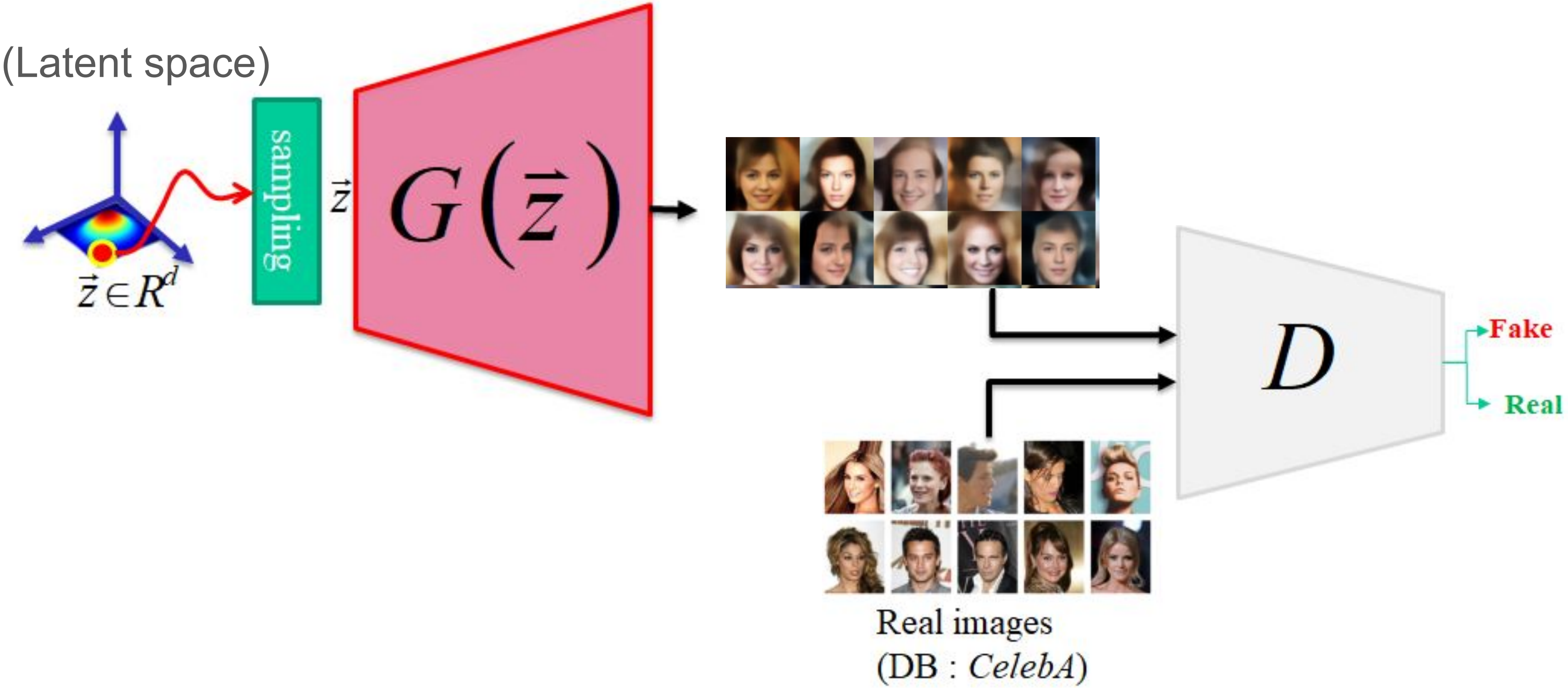
Fang Liu, etal., Deep convolutional neural network and 3D deformable approach for tissue segmentation in musculoskeletal magnetic resonance imaging. in Magnetic resonance in medicine 2018 DOI:10.1002/mrm.26841

GAN

Loss?
Annotated dataset?

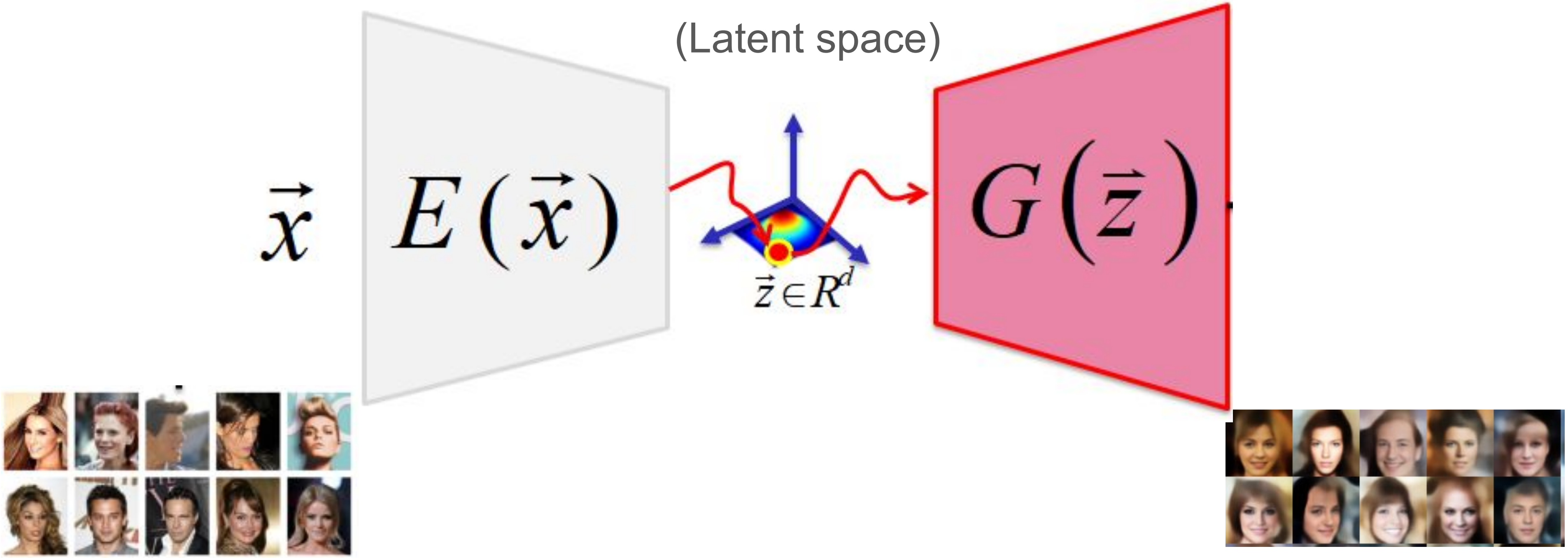


GAN



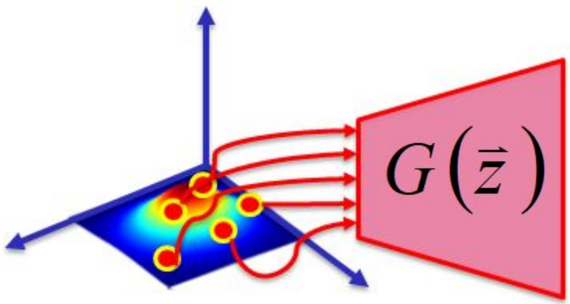
(Latent space)
$\vec{z} \in R^d$
sampling
$\vec{z}$
$G(\vec{z})$
D
Fake
Real
Real images
(DB : *CelebA*)

Autoencoders



Autoencoders (once training is over)

(Latent space)



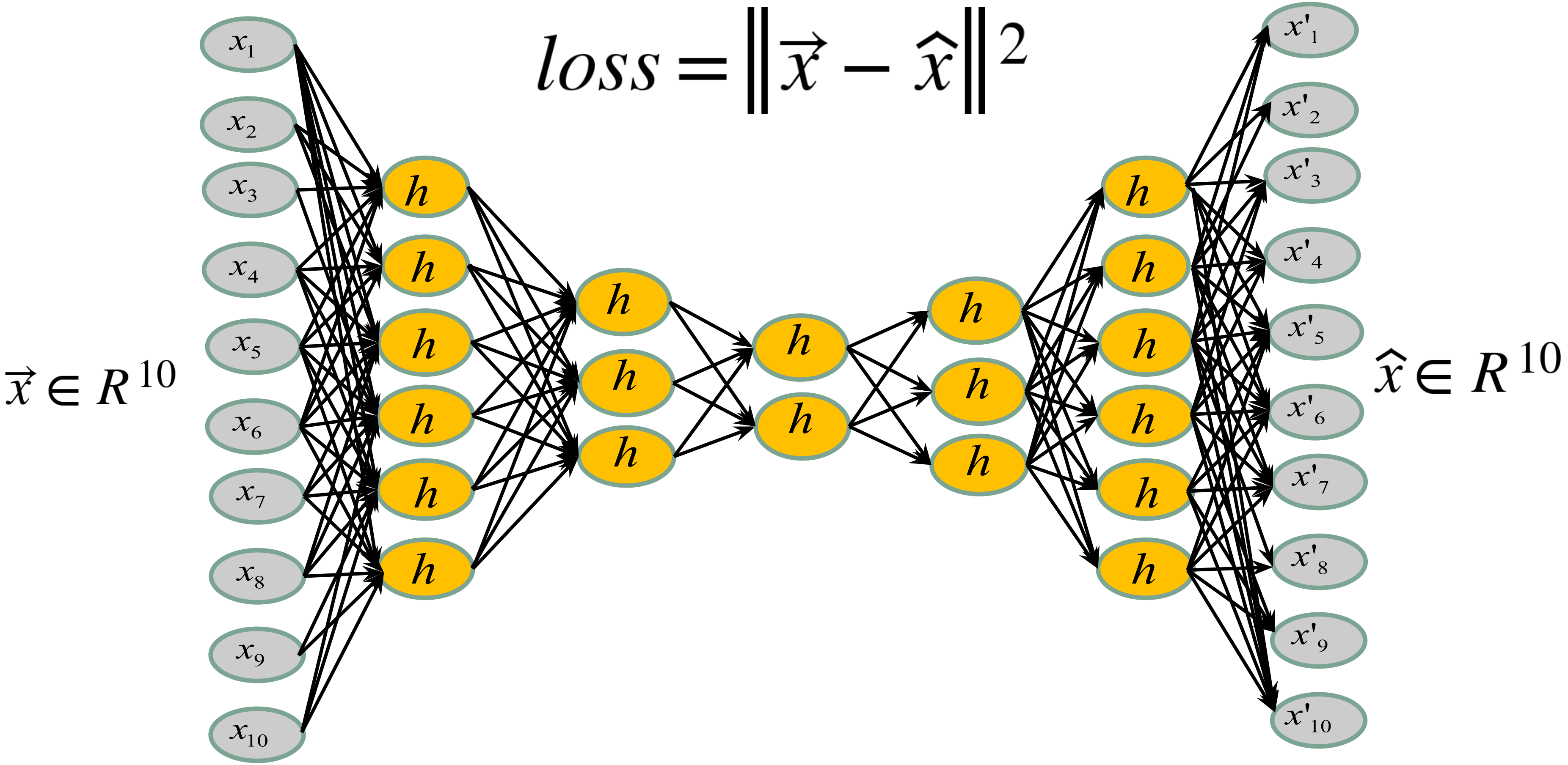
Summary

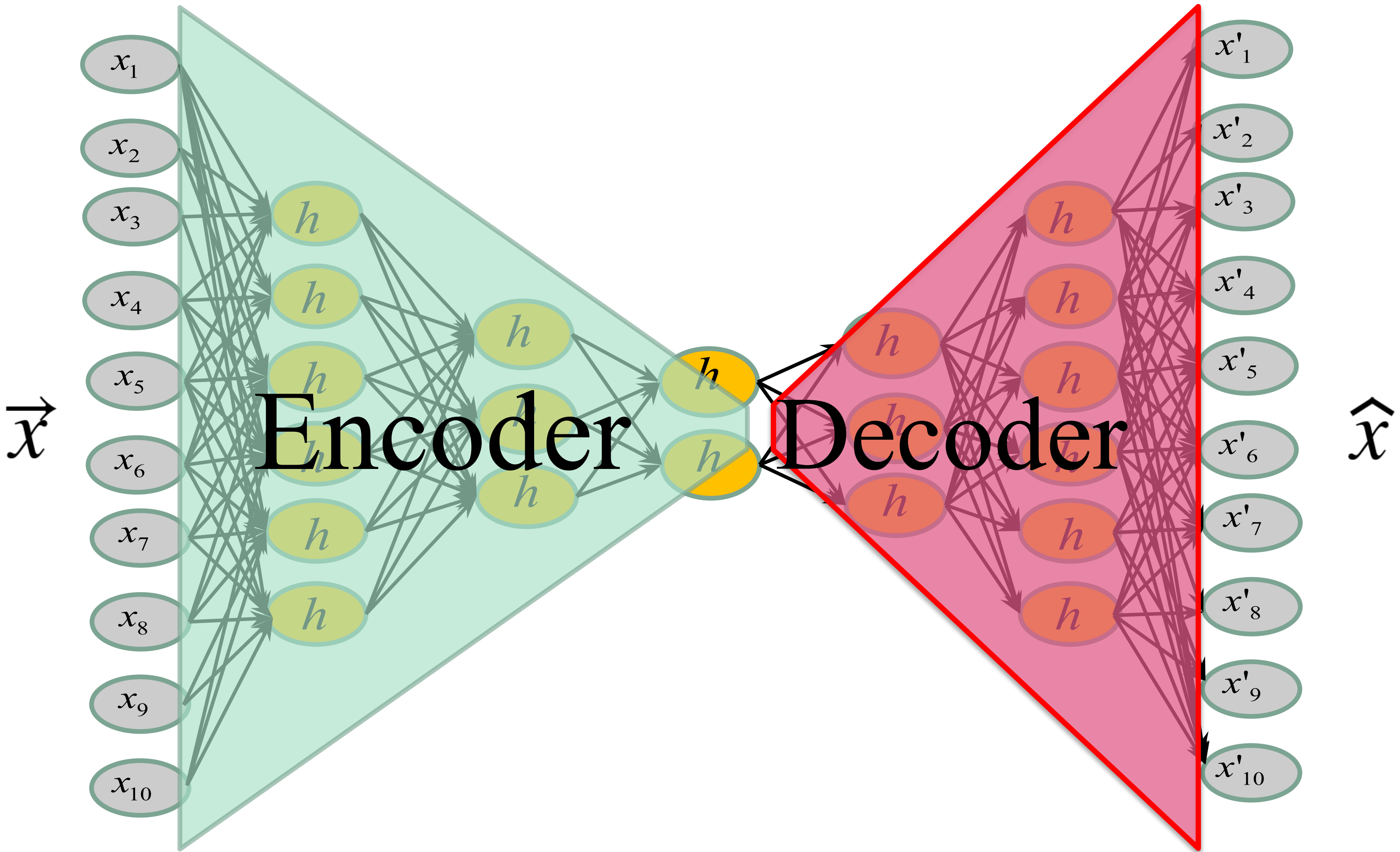
1. What are autoencoders and variational autoencoders?
2. How do they apply to MNIST (grayscale images)?
3. How do they apply to segmentation maps (ACDC cardiac labels)?

Goal : learn the latent representation of a set of data

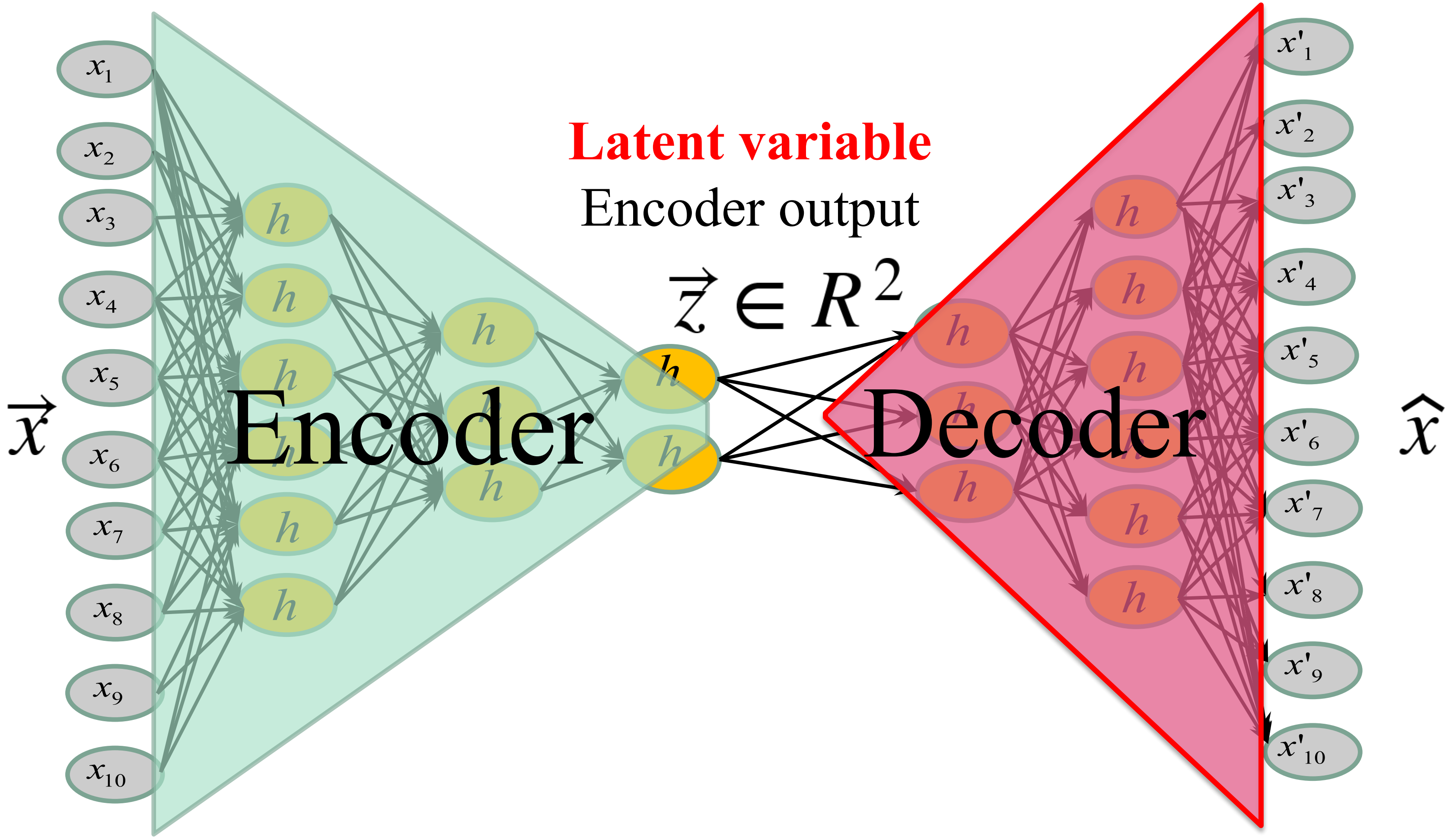
How : by training a Neural Net to output its own … input!

Note : if you are familiar with AE and VAE, you may skip the next couple of slides.

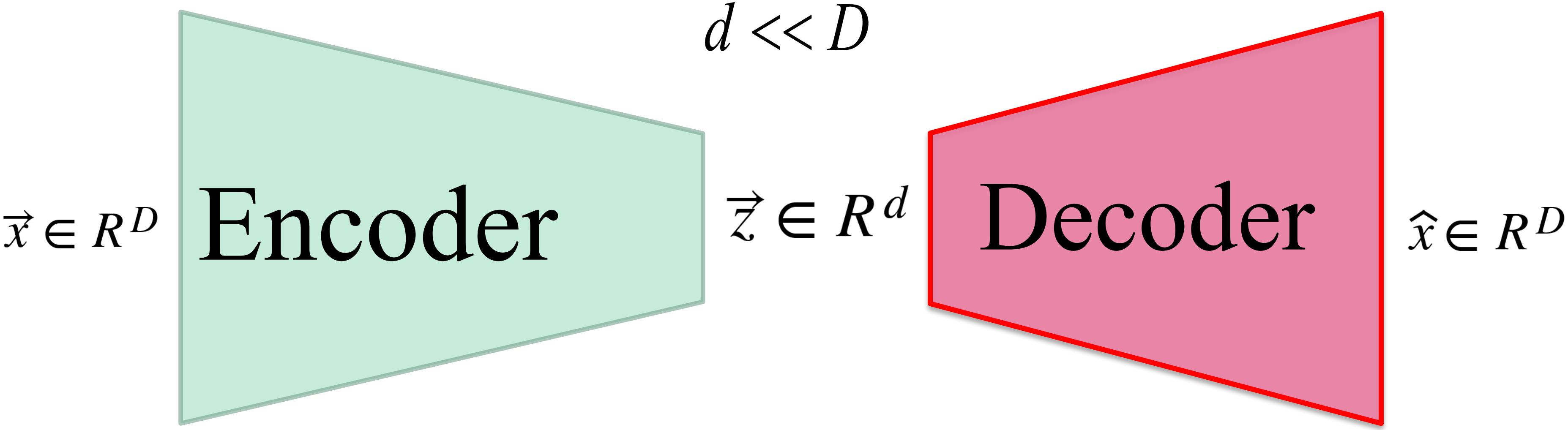

$loss = \|\vec{x} - \hat{x}\|^2$
$\vec{x} \in R^{10}$
x_1
x_2
x_3
x_4
x_5
x_6
x_7
x_8
x_9
x_{10}
h
x'_1
x'_2
x'_3
x'_4
x'_5
x'_6
x'_7
x'_8
x'_9
x'_{10}
$\hat{x} \in R^{10}$



$\vec{x}$
x_1
x_2
x_3
x_4
x_5
x_6
x_7
x_8
x_9
x_{10}
Encoder
Decoder
x'_1
x'_2
x'_3
x'_4
x'_5
x'_6
x'_7
x'_8
x'_9
x'_{10}
$\hat{x}$

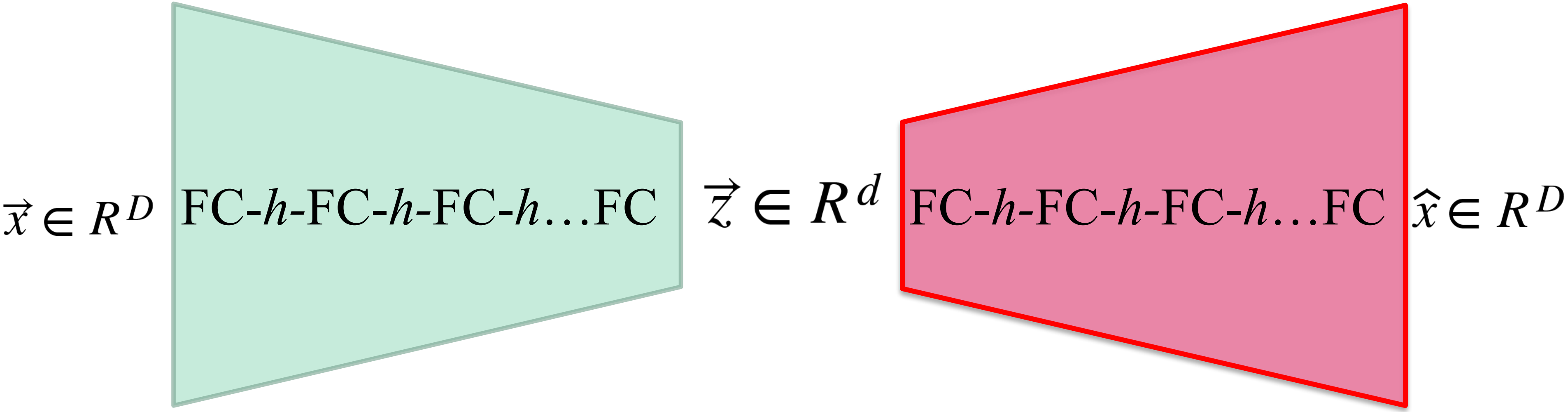


Latent variable
Encoder output
$\vec{z} \in R^2$
Encoder
Decoder
$\vec{x}$
$\hat{x}$
x_1 x_2 x_3 x_4 x_5 x_6 x_7 x_8 x_9 x_{10}
x'_1 x'_2 x'_3 x'_4 x'_5 x'_6 x'_7 x'_8 x'_9 x'_{10}

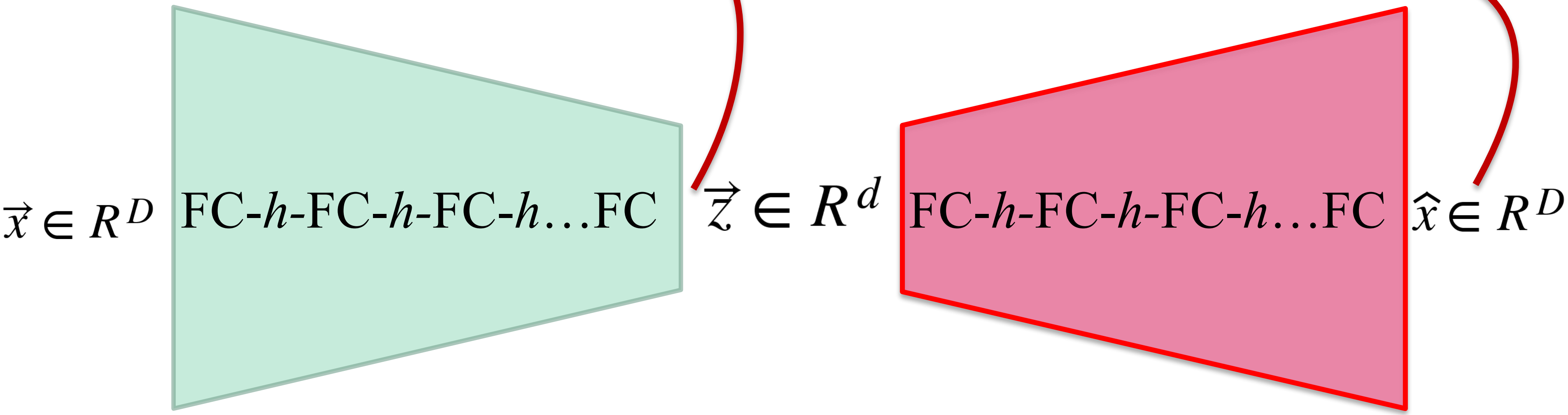

$d << D$
$\vec{x} \in R^D$
Encoder
$\vec{z} \in R^d$
Decoder
$\hat{x} \in R^D$

Fully-Connected Layers

h : activation function

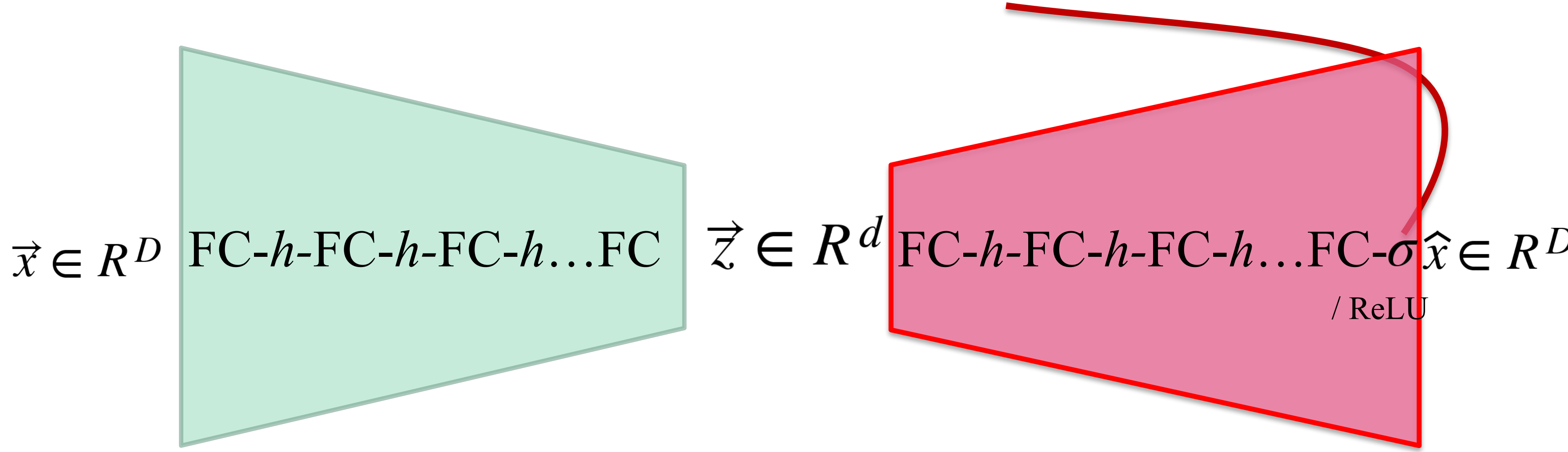


Very often...
No activation function at the output
of the encoder and the decoder



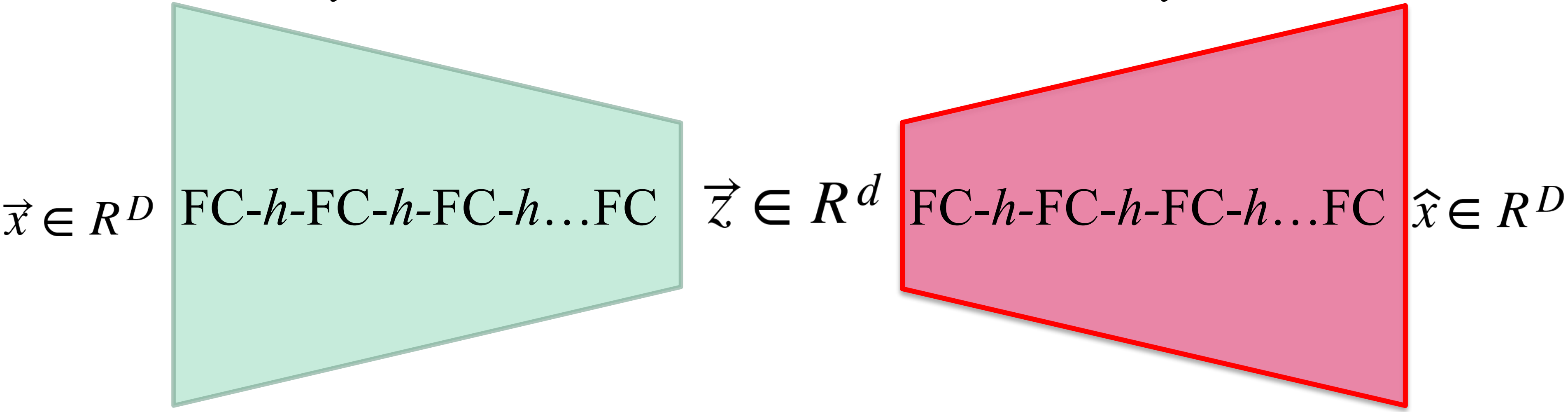
See **why?**

Sometimes **sigmoid** to predict pixel values between 0 and 1 or **ReLU** when the pixel values can be large but never negative.



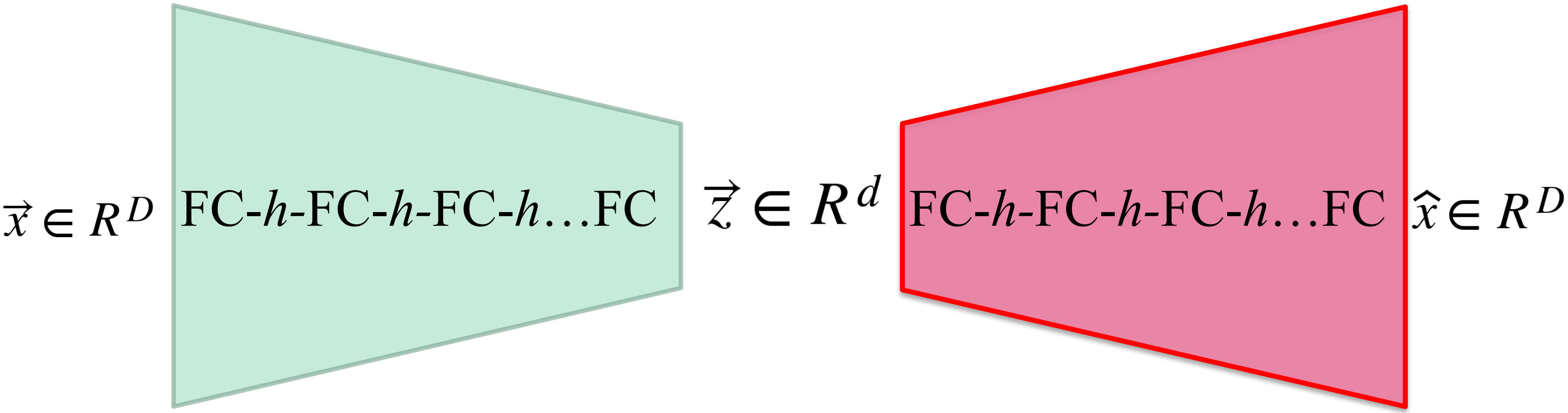
The number of neurons **Decrease (or stay the same)** from a layer to another

The number of neurons **Increase (or stay the same)** from a layer to another

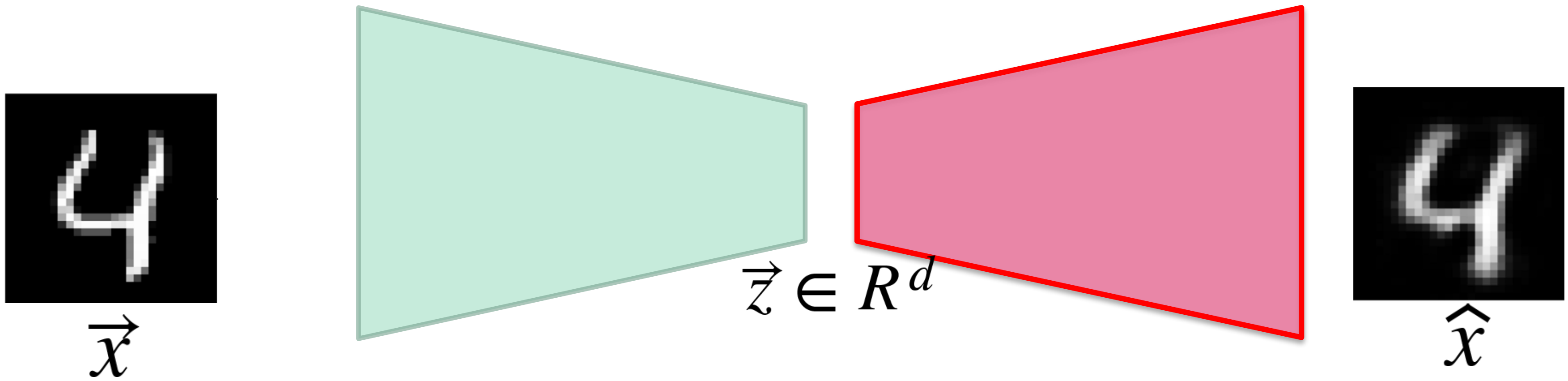


Very often...

The structure of the encoder is the **dual** of that of the decoder



Basic image-based autoencoder



Simple MNIST Autoencoder

```
class autoencoder(nn.Module):
    def __init__(self):
        super(autoencoder, self).__init__()
        self.encoder = nn.Sequential(
            nn.Linear(28 * 28, 128), nn.ReLU(True),
            nn.Linear(128, 64), nn.ReLU(True),
            nn.Linear(64, 12), nn.ReLU(True),
            nn.Linear(12, 2))
        self.decoder = nn.Sequential(
            nn.Linear(2, 12), nn.ReLU(True),
            nn.Linear(12, 64), nn.ReLU(True),
            nn.Linear(64, 128), nn.ReLU(True),
            nn.Linear(128, 28 * 28))

    def forward(self, x):
        z = self.encoder(x)
        x_prime = self.decoder(z)
        return x_prime
```

Latent space 2D

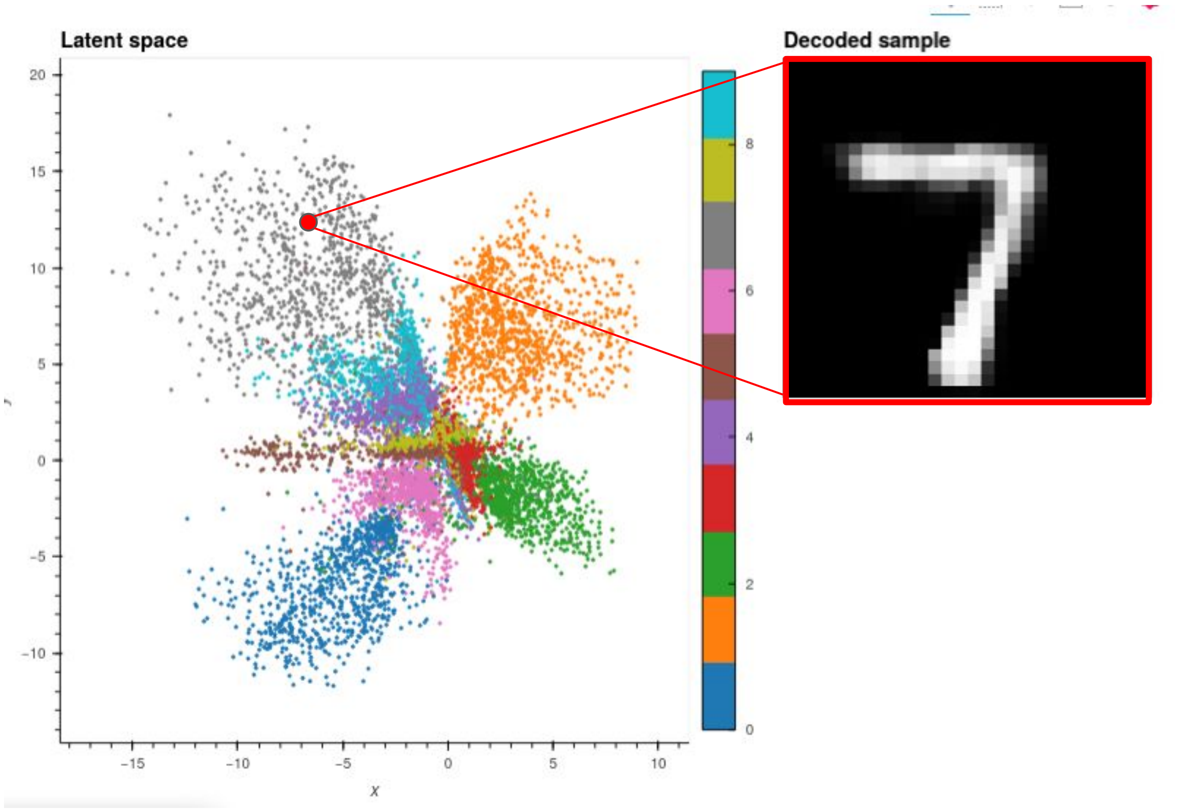
Simple MNIST Autoencoder

symmetry

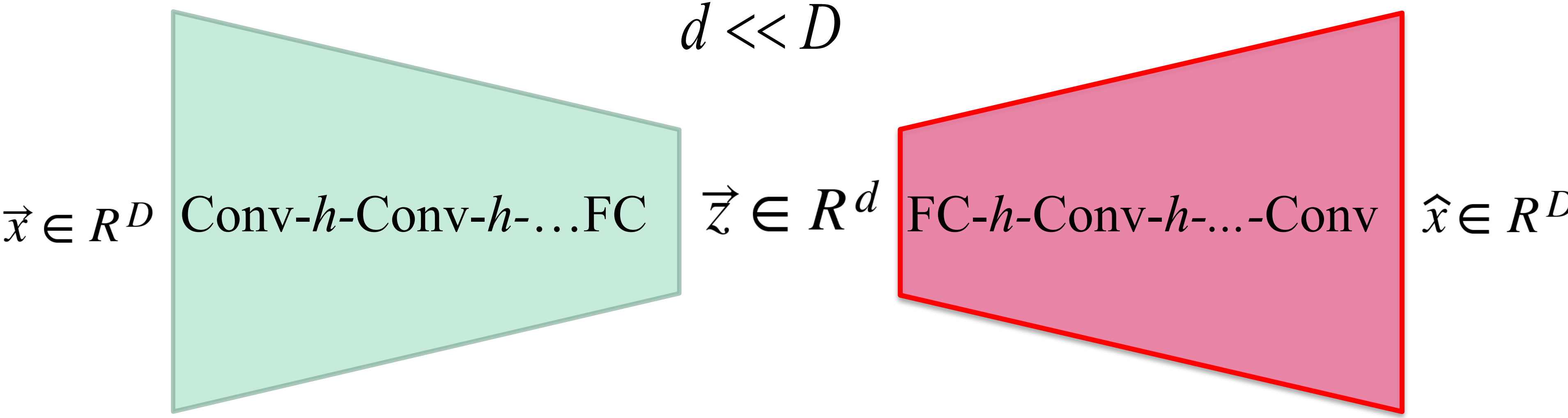
```
class autoencoder(nn.Module):
    def __init__(self):
        super(autoencoder, self).__init__()
        self.encoder = nn.Sequential(
            nn.Linear(28 * 28, 128), nn.ReLU(True),
            nn.Linear(128, 64), nn.ReLU(True),
            nn.Linear(64, 12), nn.ReLU(True),
            nn.Linear(12, 2))
        self.decoder = nn.Sequential(
            nn.Linear(2, 12), nn.ReLU(True),
            nn.Linear(12, 64), nn.ReLU(True),
            nn.Linear(64, 128), nn.ReLU(True),
            nn.Linear(128, 28 * 28))

    def forward(self, x):
        z = self.encoder(x)
        x_prime = self.decoder(z)
        return x_prime
```

MNIST latent space (for 1000 images)

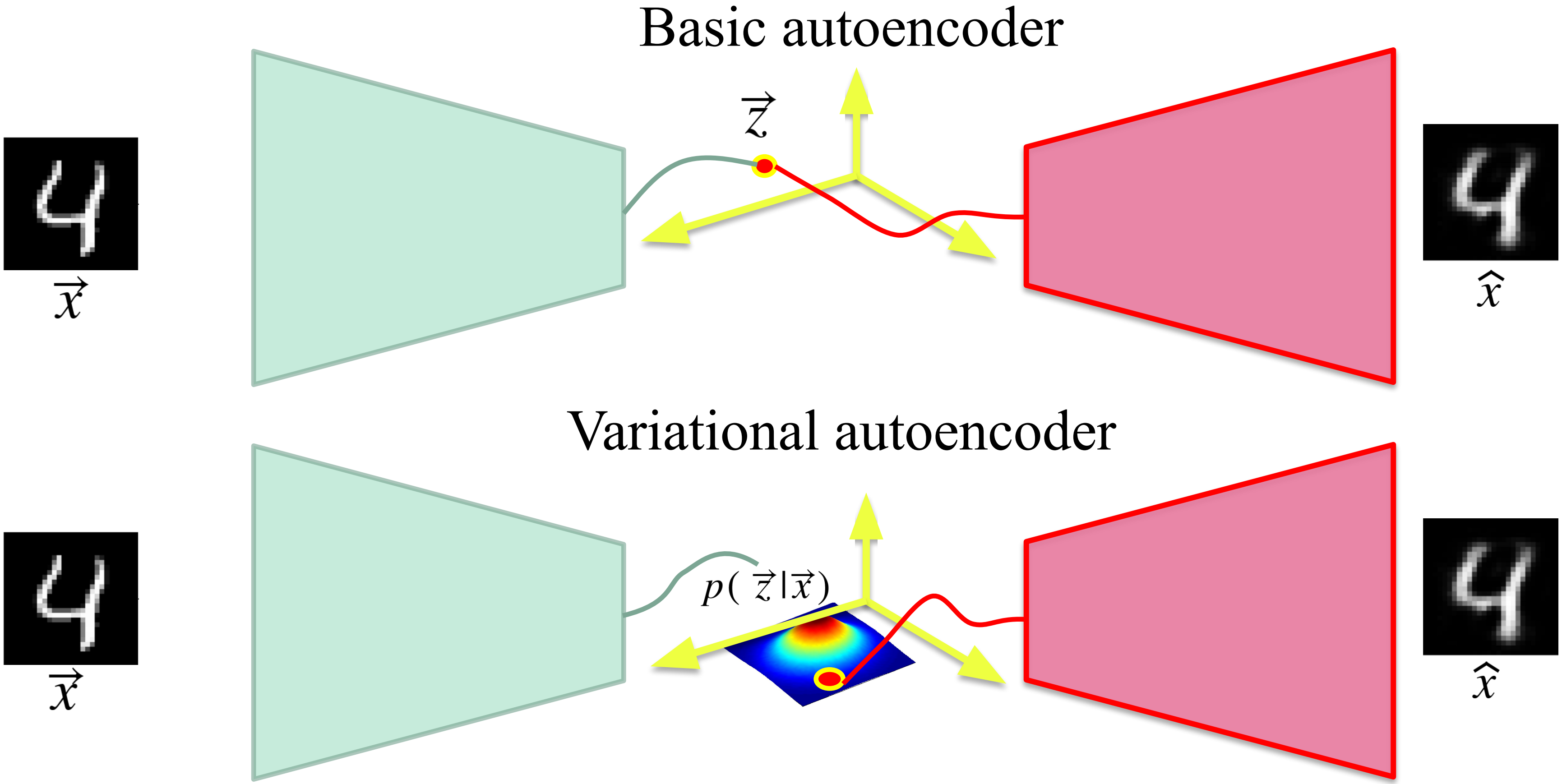
Each 2D point corresponds to an image



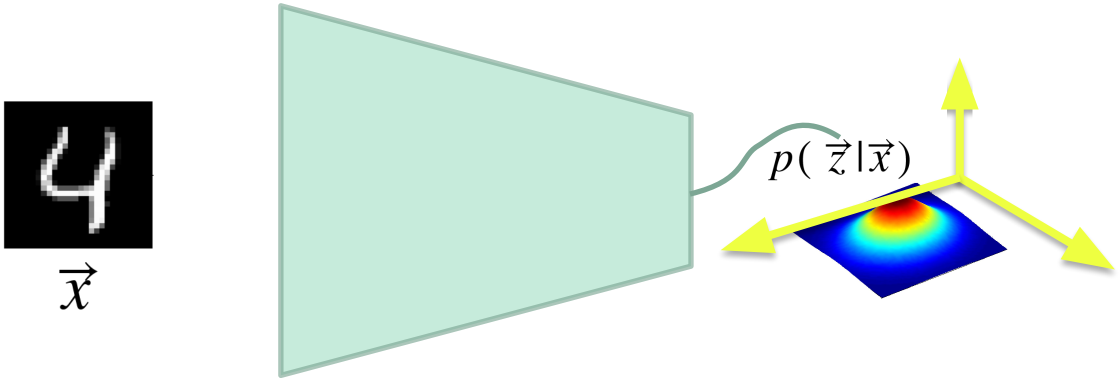
Conv layers



$d << D$
$\vec{x} \in R^D$
Conv-*h*-Conv-*h*-…FC
$\vec{z} \in R^d$
FC-*h*-Conv-*h*-...-Conv
$\hat{x} \in R^D$

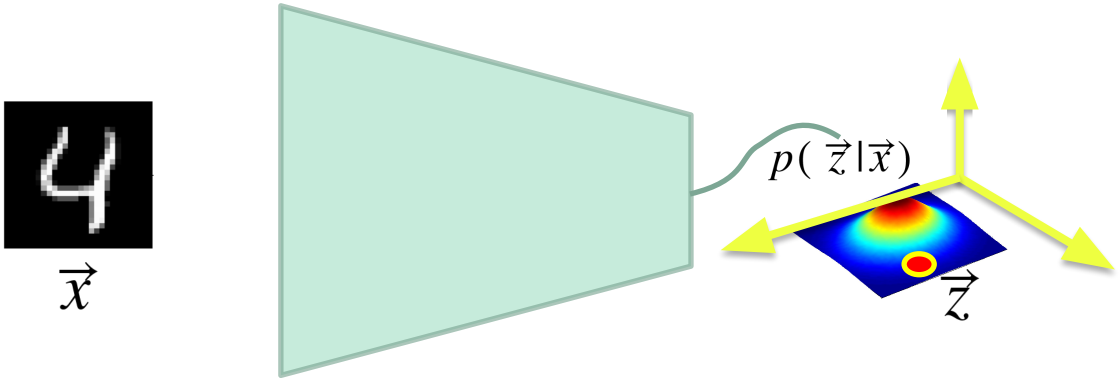

Basic autoencoder
$\vec{x}$
$\vec{z}$
$\hat{x}$
Variational autoencoder
$\vec{x}$
$p(\vec{z}|\vec{x})$
$\hat{x}$

Variational autoencoder



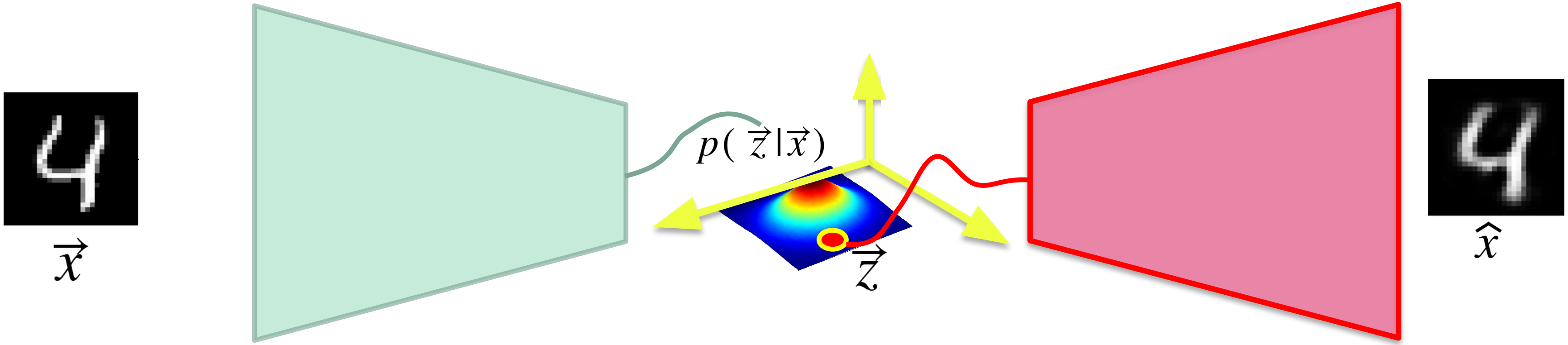
The encoder outputs a **distribution** $p(\vec{z}|\vec{x})$ and not just a **vector** $\vec{z}$

Variational autoencoder



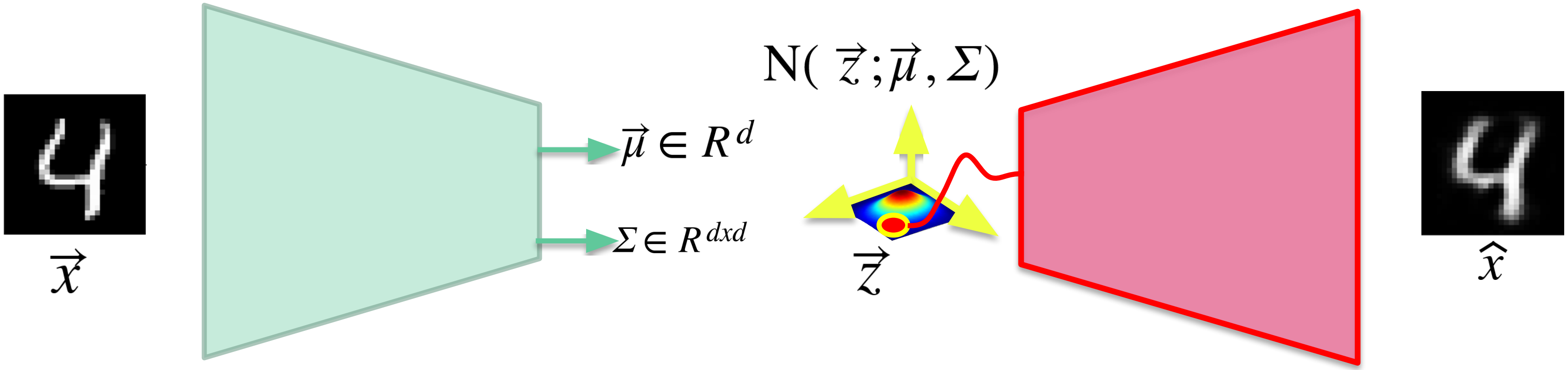
Random sample $\vec{z} \sim P(\vec{z}|\vec{x})$

Variational autoencoder

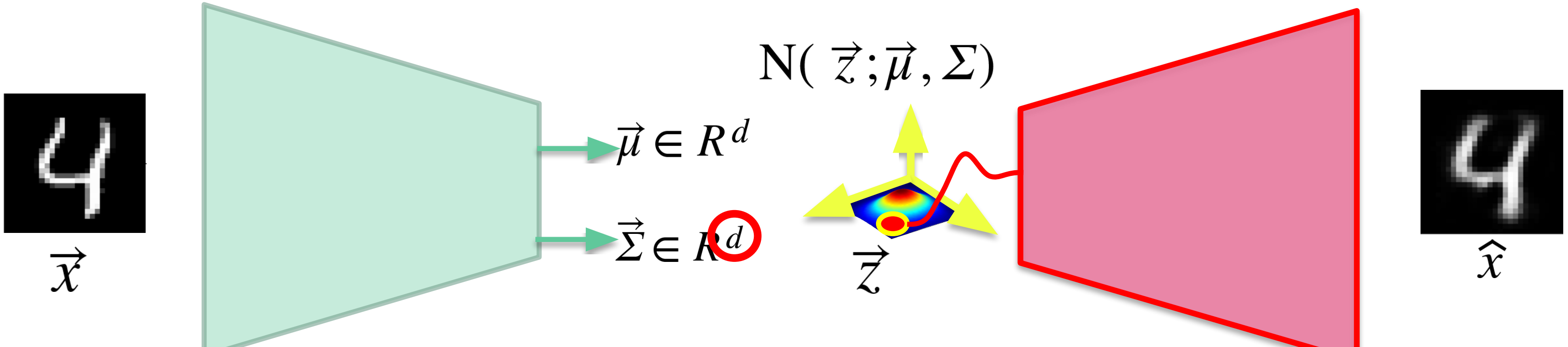


...and rebuild $\hat{x}$

$$p(\vec{z}|\vec{x}) \sim Gaussian$$



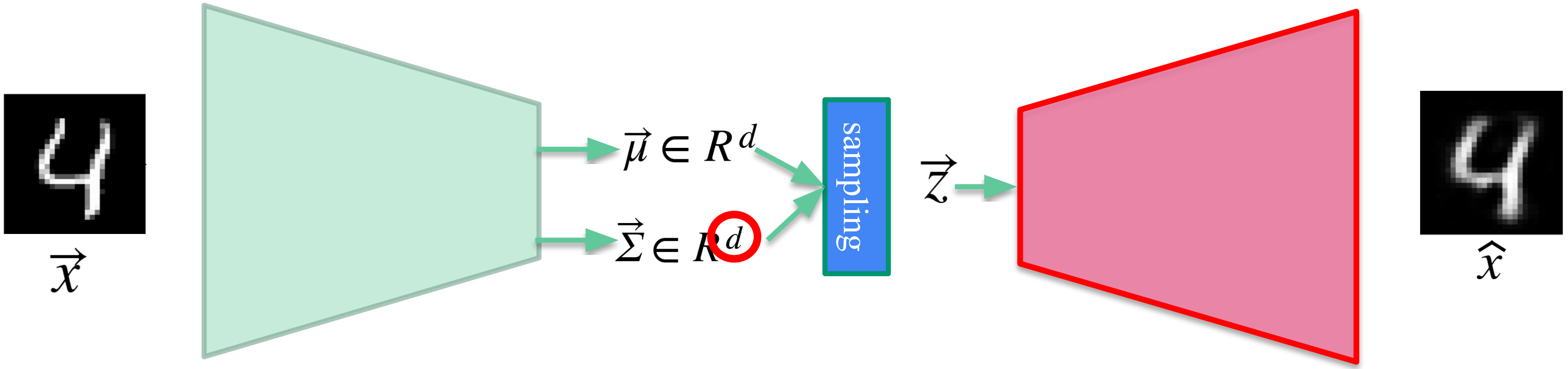
$$p(\vec{z}\,|\,\vec{x}) \sim Gaussian$$



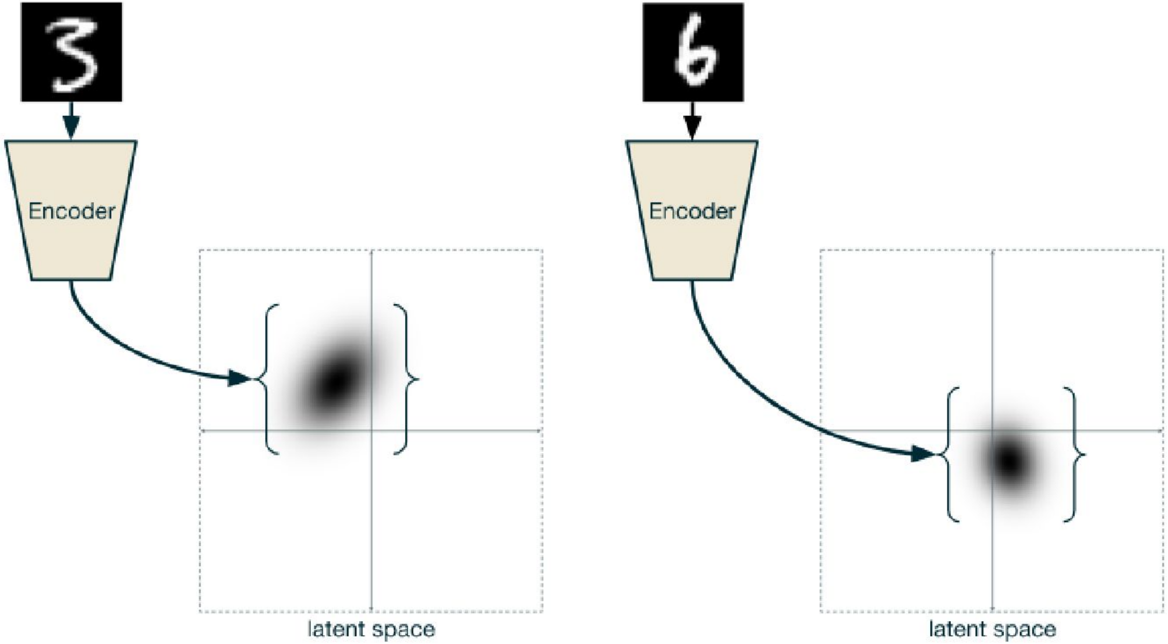
To simplify things, we assume Σ is **diagonal**

$$\Sigma = \begin{pmatrix} \sigma_1^2 & 0 & 0 & 0 \\ 0 & \sigma_1^2 & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & \sigma_d^2 \end{pmatrix}$$

Variational autoencoder

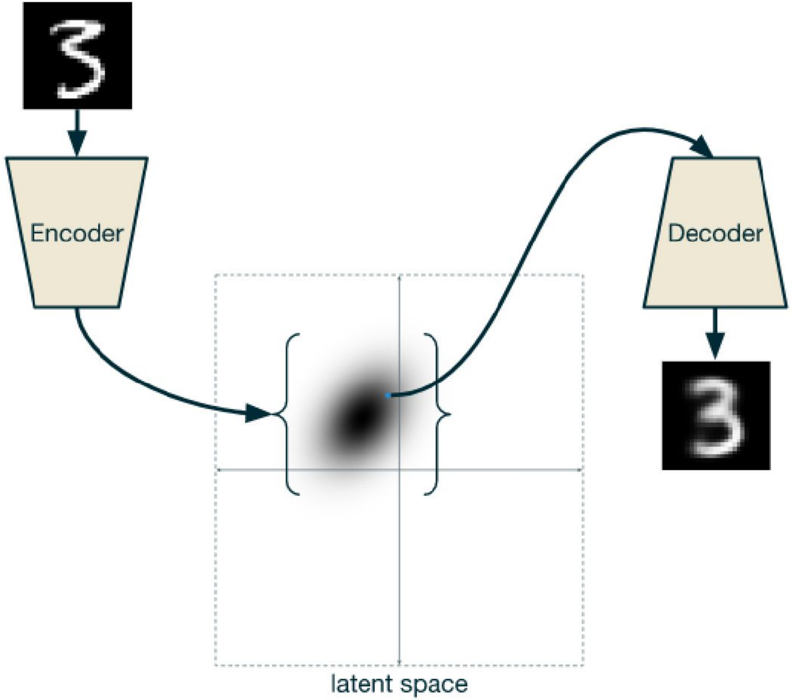


Another way of seeing things…



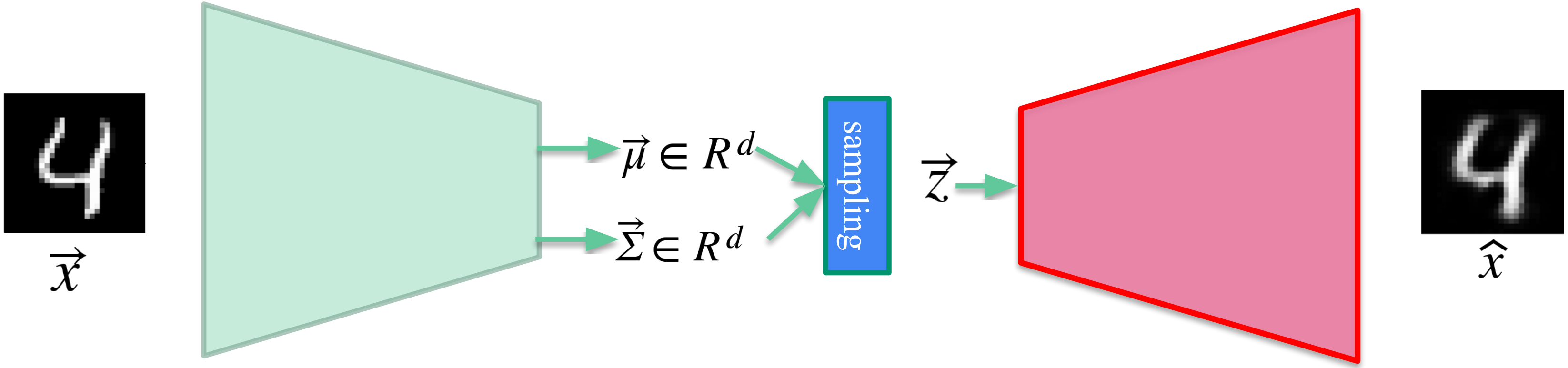
https://ijdykeman.github.io/ml/2016/12/21/cvae.html

Another way of seeing things…



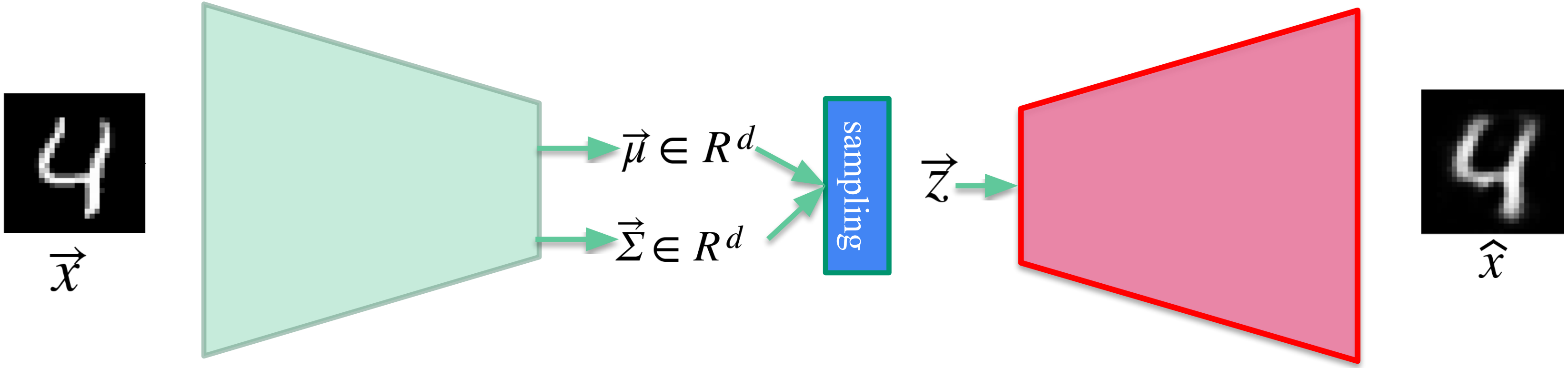
https://ijdykeman.github.io/ml/2016/12/21/cvae.html

ELBO loss : Evidence Lower Bound loss



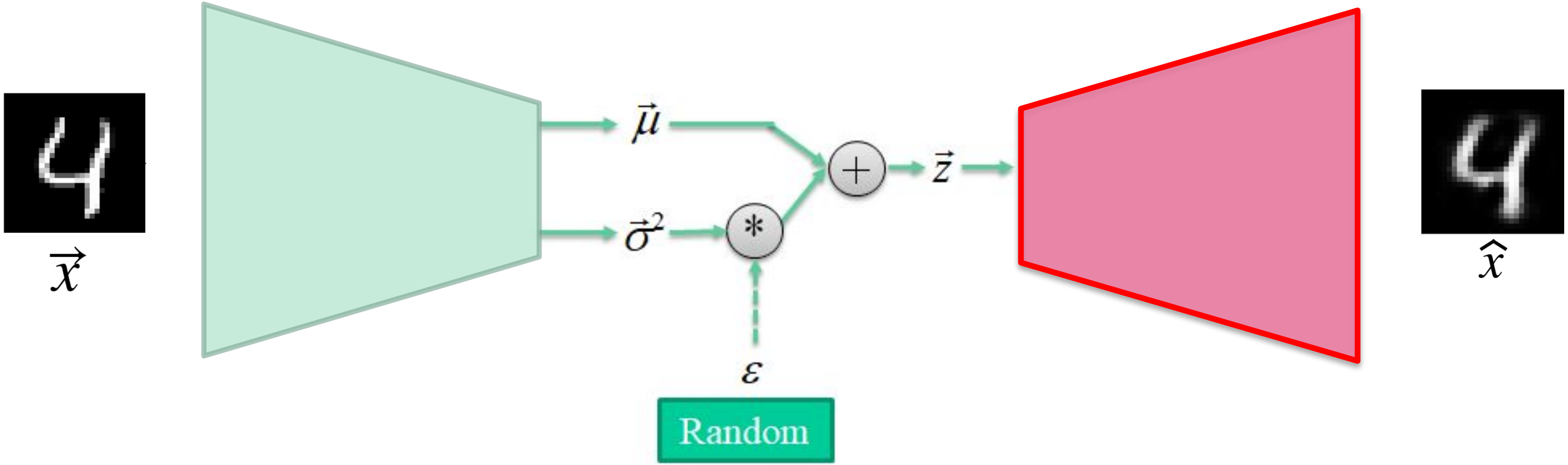
$$Loss = \underbrace{(\vec{x} - \hat{x})^{2}}_{\text{Loss decoder}} + \lambda \underbrace{KL\left(N\left(\vec{z}; \vec{0}, \vec{1}\right), N\left(\vec{z}; \vec{\mu}, \vec{\Sigma}\right)\right)}_{\text{Loss encoder}}$$

Other loss (in case the output is binary)

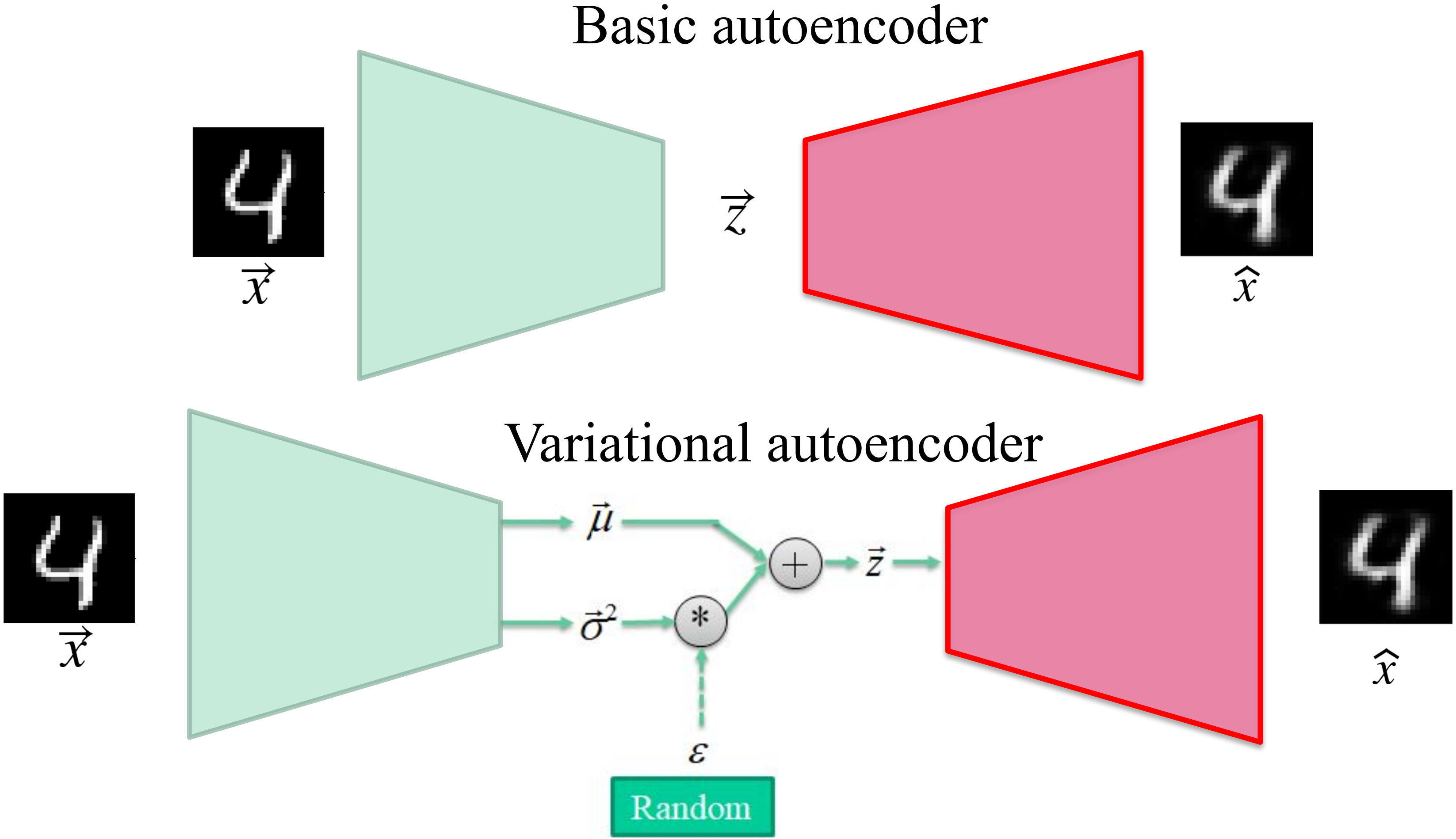


$$Loss = \underbrace{CrossEntropy(\vec{x}, \hat{x})}_{\text{Loss decoder}} + \underbrace{\lambda KL\left(N(\vec{z}; \vec{0}, \vec{1}), N(\vec{z}; \vec{\mu}, \vec{\Sigma})\right)}_{\text{Loss encoder}}$$

Reparametrization trick instead of sampling

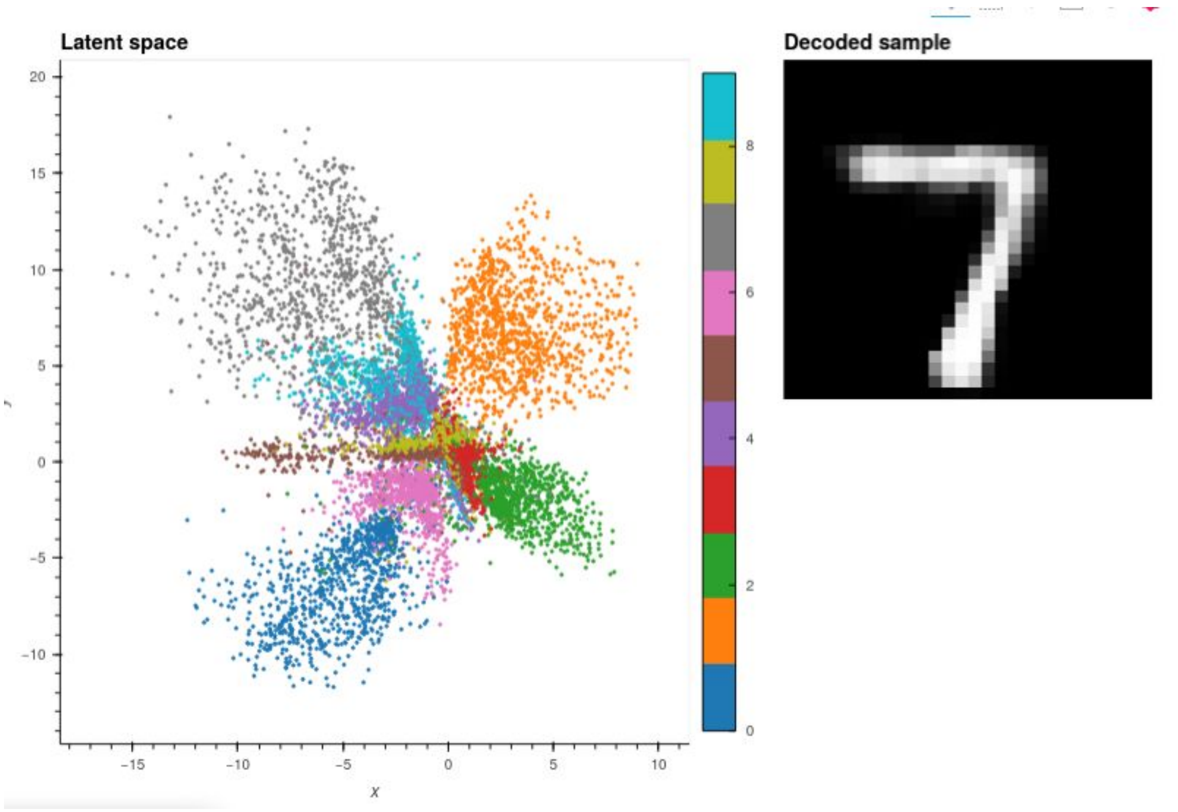


MNIST


Basic autoencoder
$\vec{x}$
$\vec{z}$
$\hat{x}$
Variational autoencoder
$\vec{x}$
$\vec{\mu}$
$\vec{\sigma}^2$
+
*
$\vec{z}$
ε
Random
$\hat{x}$

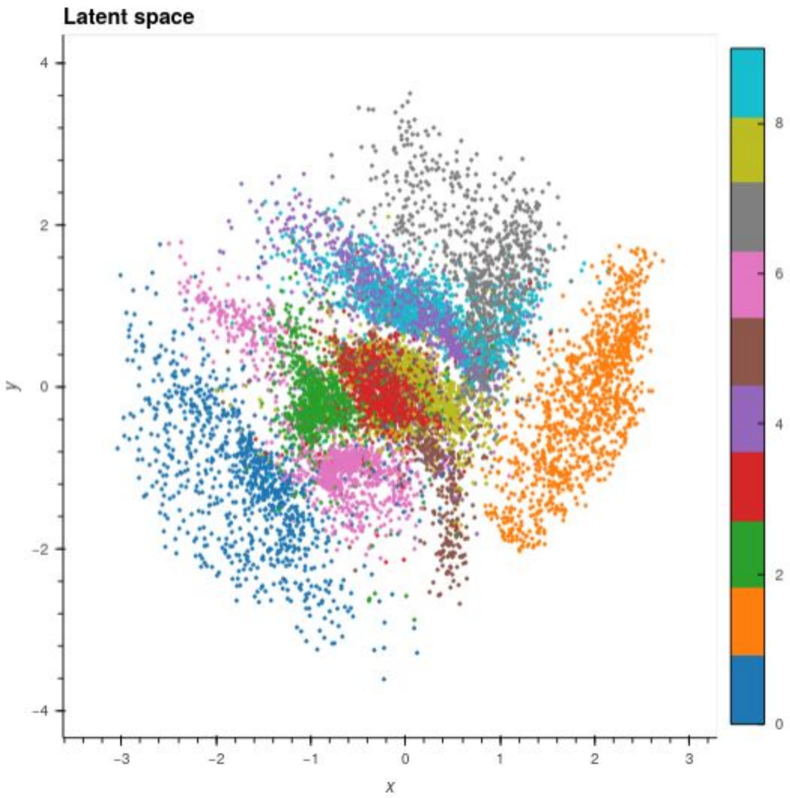
MNIST

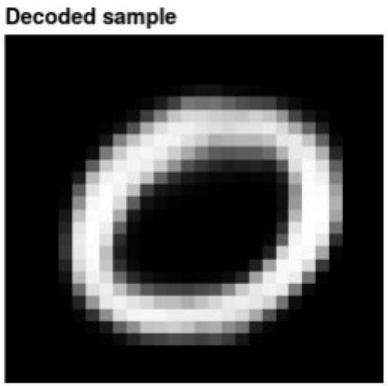
Visualize the latent space (autoencoder)



MNIST

Visualize the latent space (variational autoencoder)





MNIST

Code is in the file:

```
mnist-autoencoders.ipynb
```

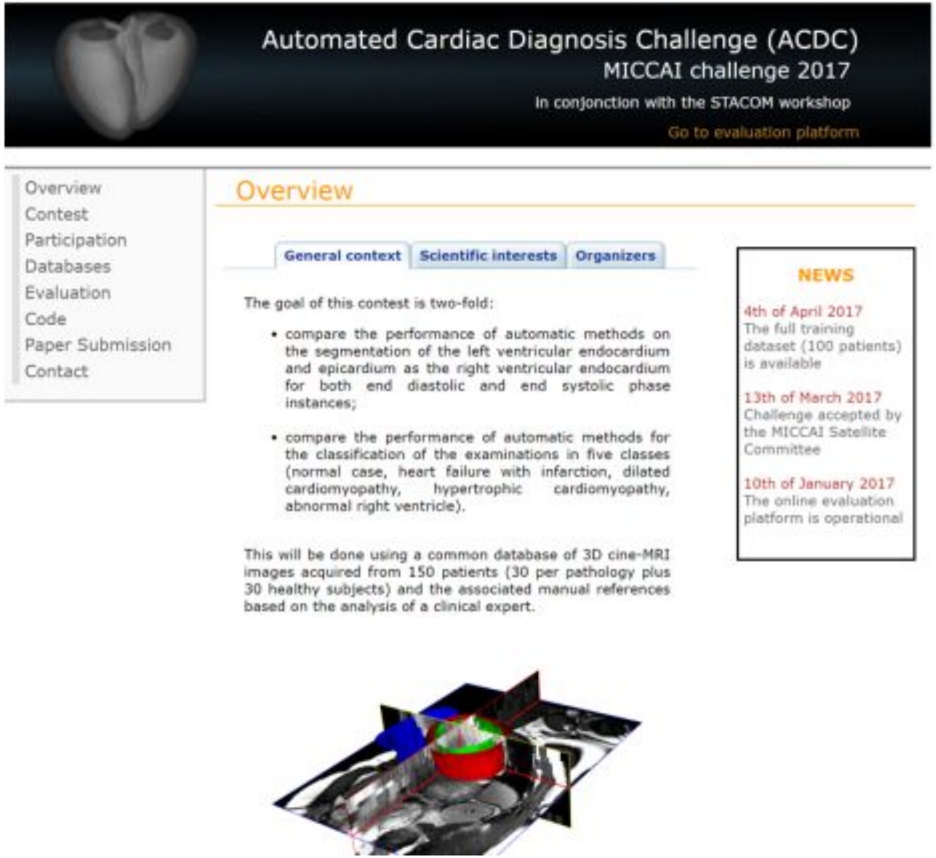
MICCAI 2017

Automated Cardiac Diagnosis Challenge

150 exams (all from different patients) divided into 5 groups:

- dilated cardiomyopathy (DCM),
- hypertrophic cardiomyopathy (HCM),
- myocardial infarction (MINF),
- abnormal right ventricle (RV)
- patients without cardiac disease (NOR).

website: ***creatis.insa-lyon.fr/Challenge/acdc***



Automated Cardiac Diagnosis Challenge (ACDC)
MICCAI challenge 2017
In conjonction with the STACOM workshop
Go to evaluation platform

- Overview
- Contest
- Participation
- Databases
- Evaluation
- Code
- Paper Submission
- Contact

Overview

General context | Scientific interests | Organizers

The goal of this contest is two-fold:

- compare the performance of automatic methods on the segmentation of the left ventricular endocardium and epicardium as the right ventricular endocardium for both end diastolic and end systolic phase instances;
- compare the performance of automatic methods for the classification of the examinations in five classes (normal case, heart failure with infarction, dilated cardiomyopathy, hypertrophic cardiomyopathy, abnormal right ventricle).

This will be done using a common database of 3D cine-MRI images acquired from 150 patients (30 per pathology plus 30 healthy subjects) and the associated manual references based on the analysis of a clinical expert.

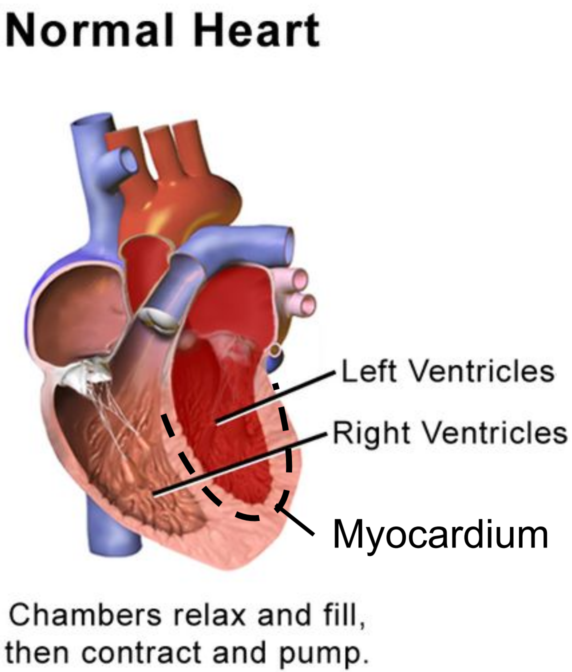
NEWS

4th of April 2017
The full training dataset (100 patients) is available

13th of March 2017
Challenge accepted by the MICCAI Satellite Committee

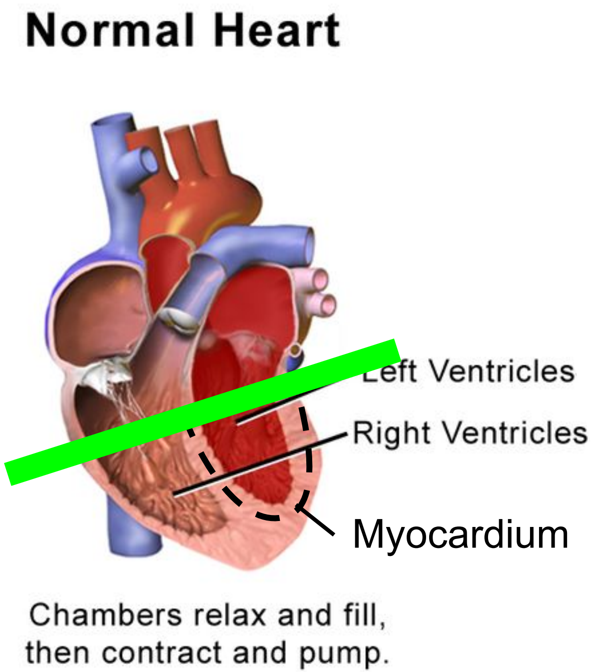
10th of January 2017
The online evaluation platform is operational

Cardiac anatomy

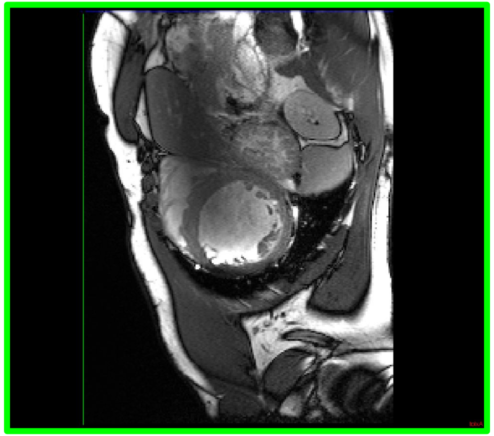


https://commons.wikimedia.org/

Cardiac anatomy

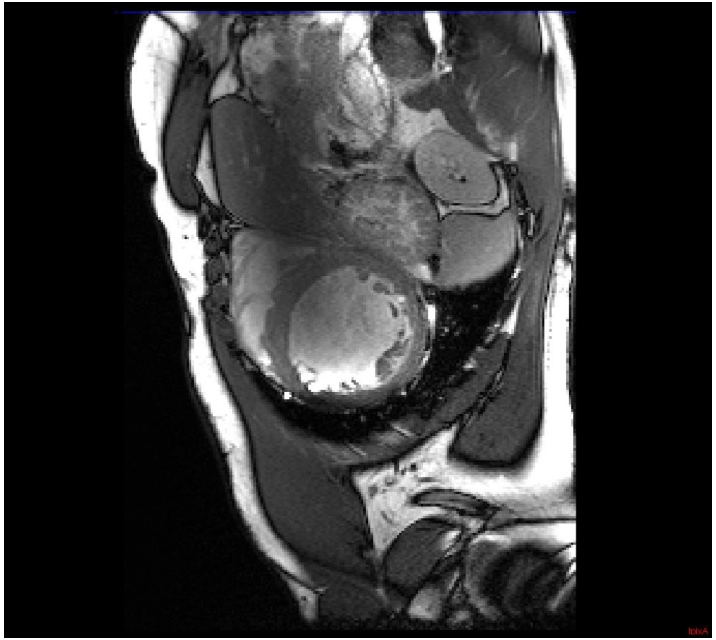


Cross-section : **short axis view**

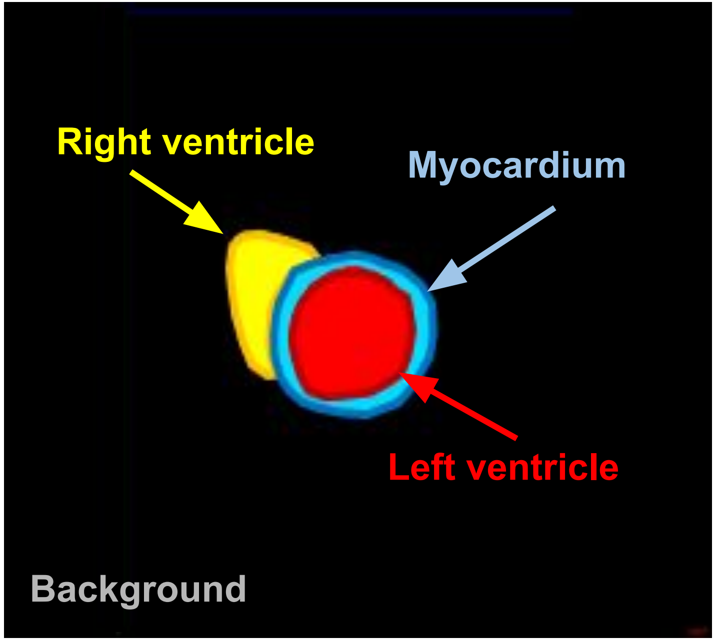


https://commons.wikimedia.org/

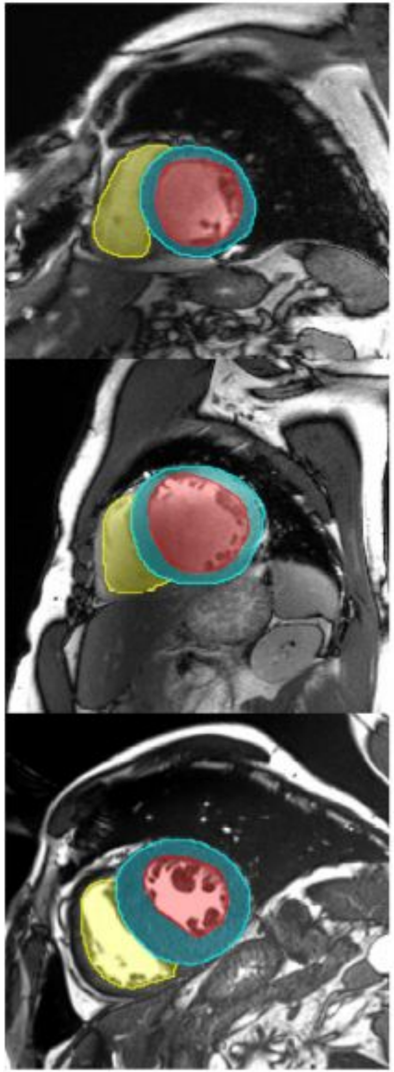
Short axis cardiac MRI



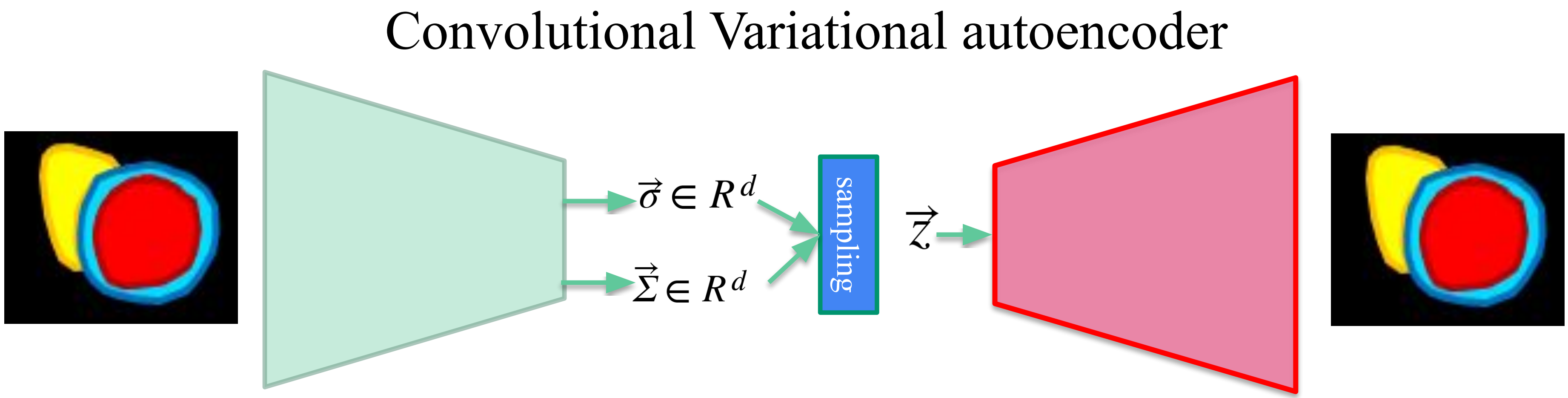
Short axis segmentation map



Other examples:



ACDC


Convolutional Variational autoencoder
$\vec{\sigma} \in R^d$
$\vec{\Sigma} \in R^d$
sampling
$\vec{z}$

ACDC

Code is in the file:

`cardiac-mri-autoencoders.ipynb`

Thank you